CCR/XVA Fundamentals

APRIL 2026

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Counterparty Credit Risk (CCR)

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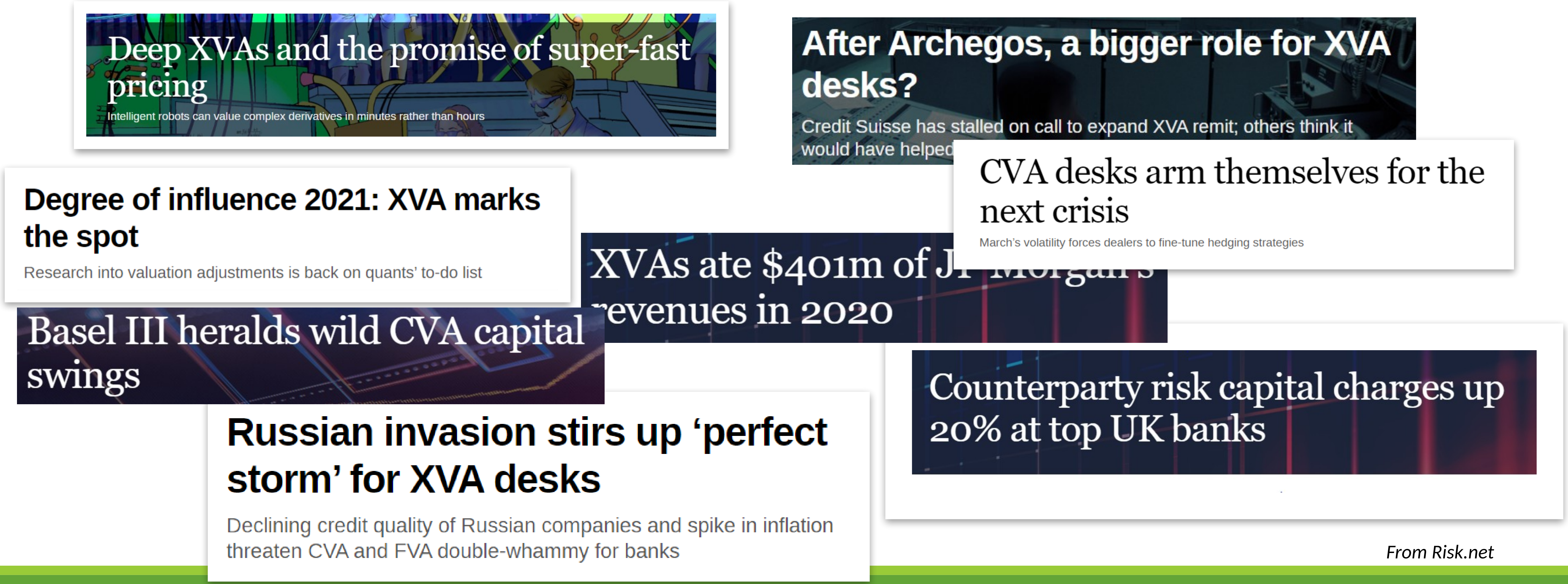
Contents

Introduction: Motivation

Counterparty Credit Risk & CVA are a key focus in the derivatives market



Deep XVAs and the promise of super-fast pricing

Intelligent robots can value complex derivatives in minutes rather than hours

After Archegos, a bigger role for XVA desks?

Credit Suisse has stalled on call to expand XVA remit; others think it would have helped

CVA desks arm themselves for the next crisis

March's volatility forces dealers to fine-tune hedging strategies

Degree of influence 2021: XVA marks the spot

Research into valuation adjustments is back on quants' to-do list

XVAs ate $401m of J

revenues in 2020

Basel III heralds wild CVA capital swings

Counterparty risk capital charges up 20% at top UK banks

Russian invasion stirs up 'perfect storm' for XVA desks

Declining credit quality of Russian companies and spike in inflation threaten CVA and FVA double-whammy for banks

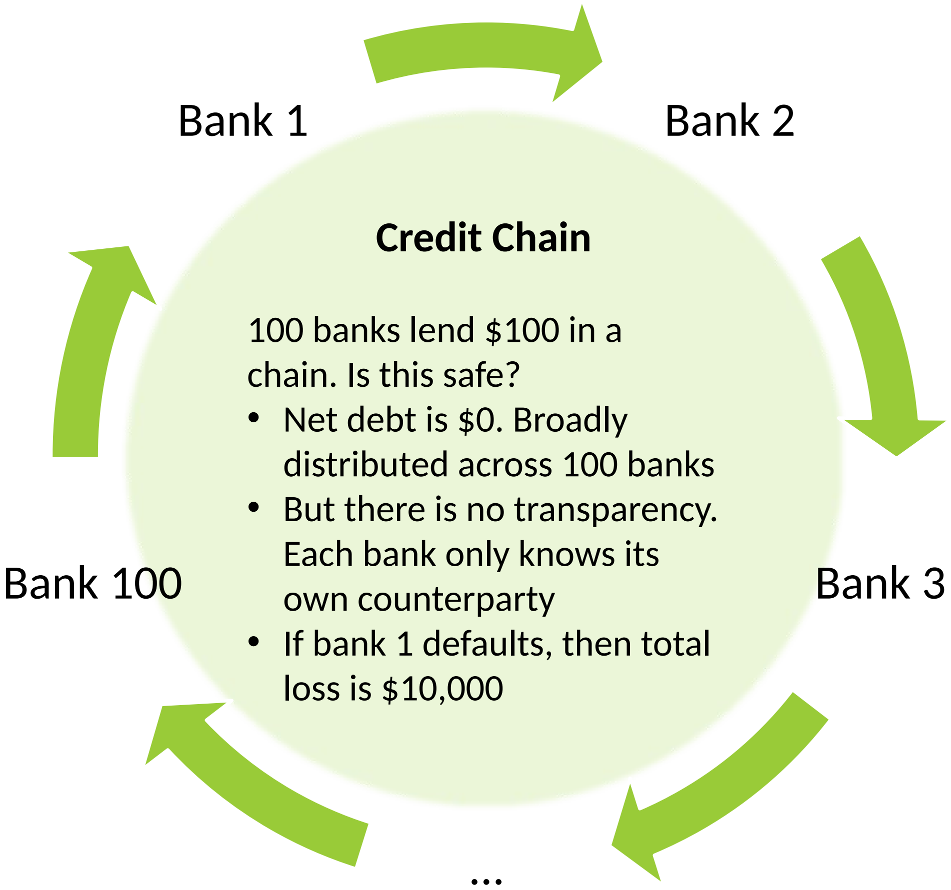
From Risk.net

Introduction: Definitions

What is Counterparty Credit Risk (CCR)?

> *…the risk that the counterparty to a transaction could default before the final settlement of the transaction's cash flows. [Basel CRE50, 50.1]*

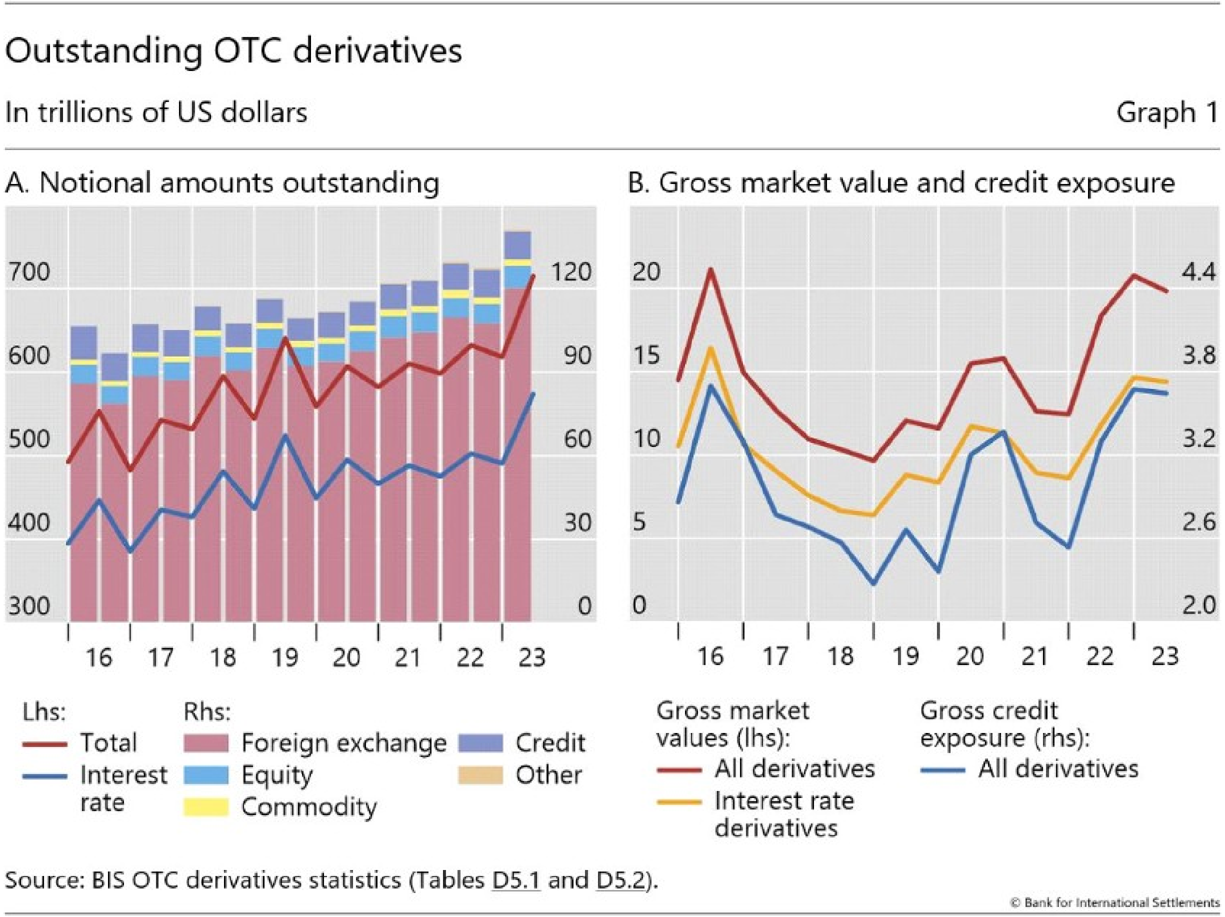
- Applies to **derivatives** transactions.
  - Primarily in the **Over-the-Counter** (OTC) market
- Why is this special?
  - **Bi-lateral** credit risk
  - **Uncertain** exposure: Market contingent credit risk
  - **Contagion**: Back-to-back transactions entail counterparty risk
- Drivers:
  - Default probability, market volatility, portfolio size, maturity



Introduction
The OTC Derivatives Market

Key Participants (Counterparty):

- Banks
- Hedge funds
- Other financial institutions
  - Pension funds
  - Broker-dealers
- Corporate Clients
- Governments and government agencies
- Regulators



Introduction: Risk Management

How is Counterparty Credit Risk managed?

Risk Limits

- **Credit risk officers** monitor individual counterparties
- Metrics
  - **Potential future exposure**
  - Stress losses

Collateral

- Hold high quality assets to offset credit risk
- Collateral types
  - **Variation Margin**
  - **Initial Margin**
- Central Counterparties

Pricing

- **CVA trading team** hedges credit risk
- CVA:
  - **Credit Valuation Adjustment**
  - Cost of hedging
  - Charged to clients

Capital

- Regulators set minimum **capital requirements** to protect against credit risk
- **“Basel Rules”**
  - Default RWA
  - CVA RWA

Foundations: Outline

Exposure

- How much could we lose upon counterparty default?
  - Expected Exposure Profile, Potential Future Exposure

Examples

- What does the expected exposure profile look like?
  - Forwards, Swaps

Mitigation

- How can we reduce the exposure?
  - Netting, Collateral

Foundations: Forward MTM

MTM and Future MTM

- Initial MTM: $V(0)$
  - Future MTM: $V(t)$
  - $V(t)$ is the risk-free value (mark-to-market) of the trade at time t.
  - For $t > 0$, $V(t)$ is not known and hence is a random variable.

Marginal Deals

Trade Name	Credit End Date	Recalc MTM
trade2y_Rec 0 USD / Pay -1,000,000 US...	04/23/2026	902,938.57
trade2y_Rec 1,000,000 USD / Pay 0 USD...	04/23/2025	946,880.24

Portfolio with two trades with Fixed Cash Flow:

Date of Pricing is 23rd April 2025

- Trade 1 : 1,000,000 USD; Maturity = 1Y (23rd April 2026)
- Trade 2 : 1,000,000 USD ; Maturity = 2Y (23rd April 2027)
- Portfolio Value = MTM(Trade 1) + MTM(Trade 2)
- Portfolio Initial MTM = 1,849,818 USD-? Why not 2,000,000 ?
- Future MTM(24 April 2026) ~1,00000

$$MTM = V(0) = \mathbb{E}_0^Q[D_0(T)\Phi(T)]$$

$D_0(T)$ = Discount Factor

$\Phi(T)$ = Payoff : e.g. 1,000,00 USD

$$Future\ MTM = V(t) = \mathbb{E}_t^Q[D_t(T)\Phi(T)]$$

-> Scenario dependent. Because we don't know the value of asset at future date "t"

Foundations: Forward MTM

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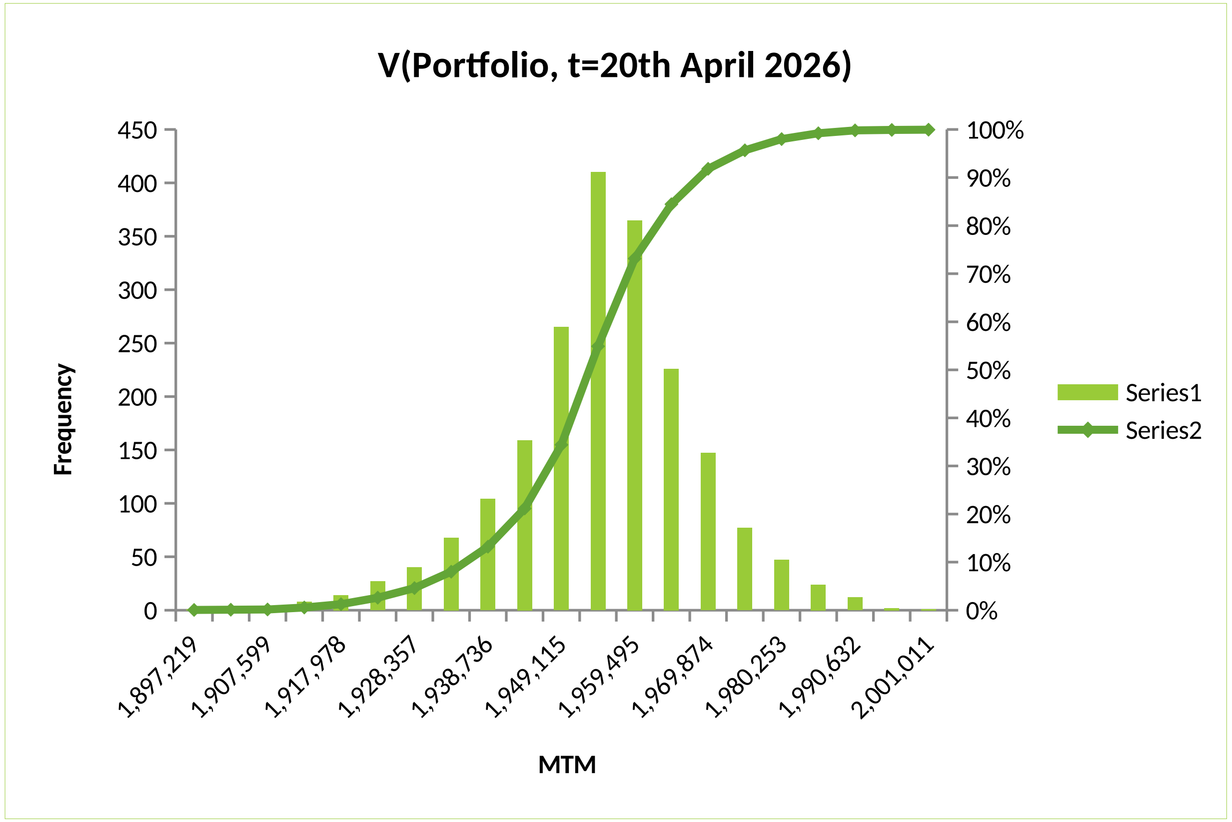
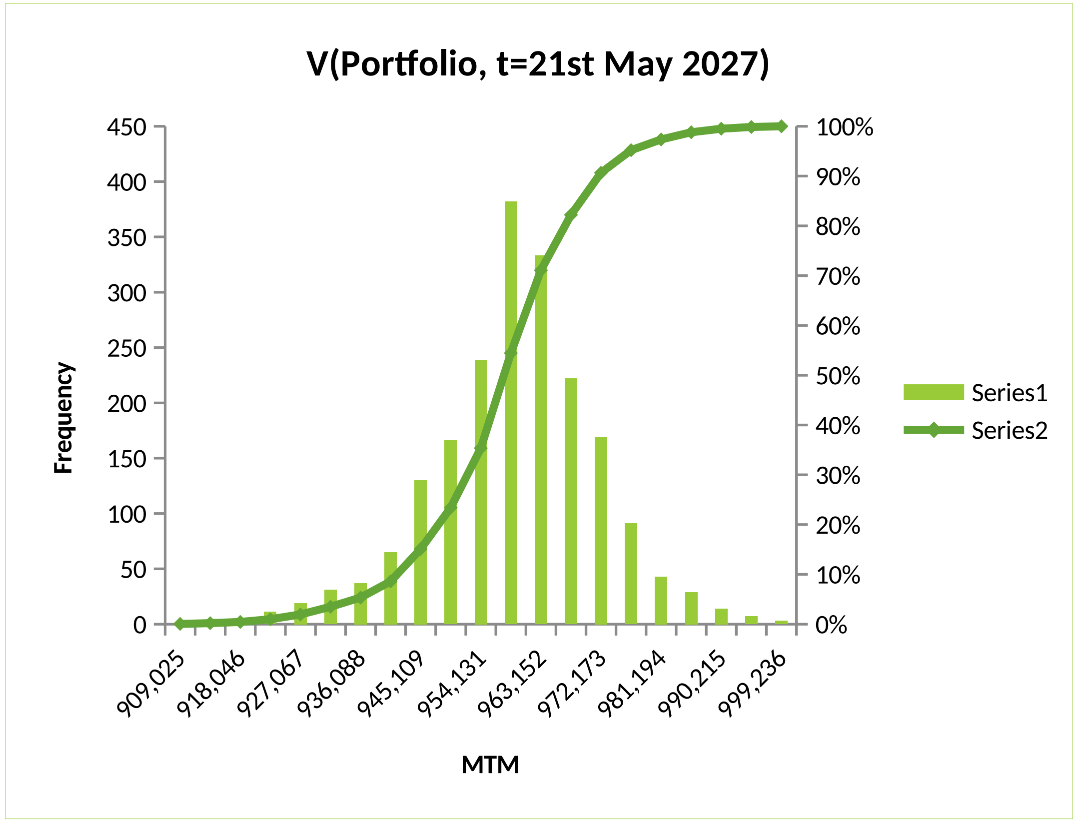
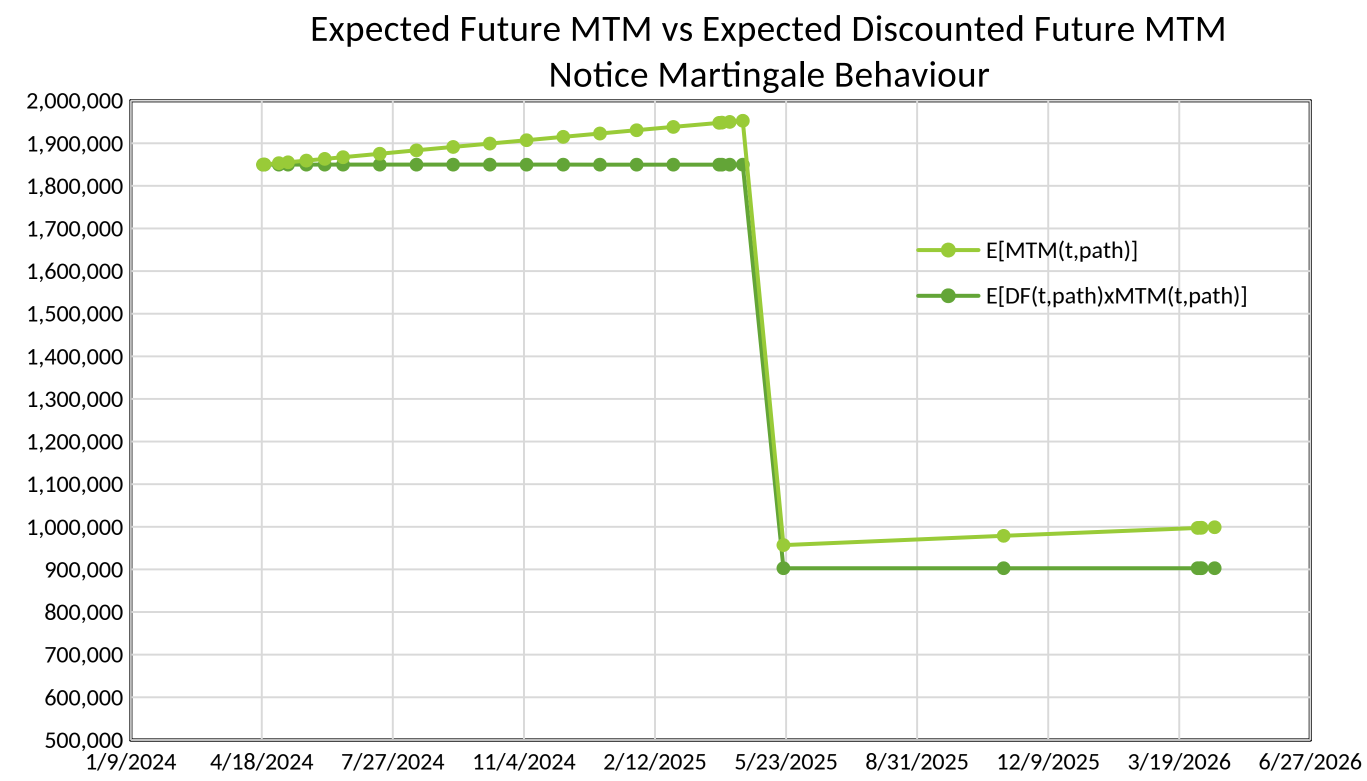




Foundations: Forward MTM



$$\mathbb{E}_0^Q[D_0(\tau)V_{RF}(\tau)] = \mathbb{E}_0^Q\left[D_0(\tau)\mathbb{E}_\tau^Q[D_\tau(T)\Phi(T)]\right]$$

$$= \mathbb{E}_0^Q\left[\mathbb{E}_\tau^Q[D_0(\tau)D_\tau(T)\Phi(T)]\right]$$

$$= \mathbb{E}_0^Q[D_0(T)\Phi(T)]$$

$$=V_{RF}(0)$$

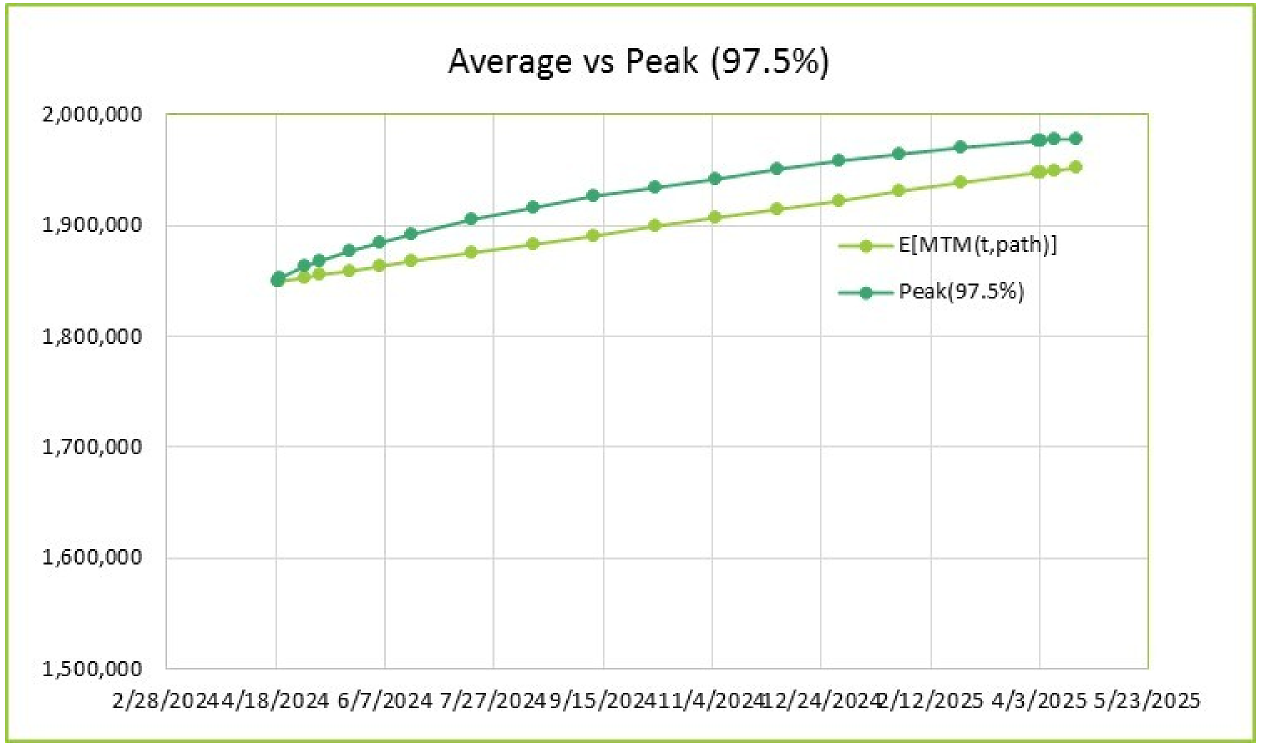
=Initial MTM

Foundations: Forward MTM

.

Expected Exposure Profile

- Curve of expected exposure for all future t as of today t_0:
  - $\mathrm{EE}(t) = \mathbb{E}_0^*[E(t)|t=\tau] = \mathbb{E}_0^*[V(t)^+|t=\tau]$
- **Conditional on counterparty defaulting at time $t = \tau$**
- If exposure is independent of default:
  - $\mathrm{EE}(t) = \mathbb{E}_0^*[V(t)^+]$



Peak (97.5%) : Exposure at default Metrics used for Risk Management

Foundations: Peak 97.5%

Marginal Deals

Trade Name	Maturity Date	Product Type	Pay CCY	Receive CCY
fxfwdUSDINR_Rec 1,000,000 USD / Pay 84,000,000 INR (M...	04/22/2025	FX FWD	INR	USD

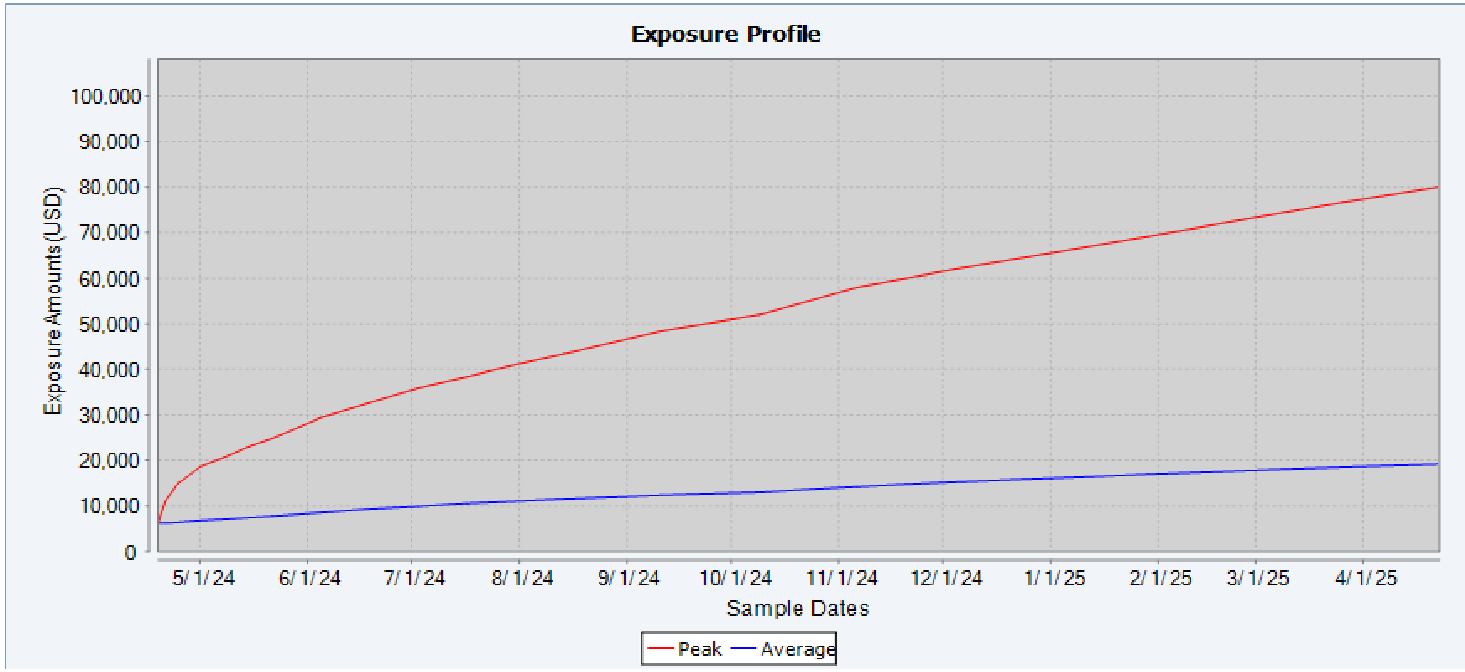
Peak (97.5%) : Metrics used for Risk Management.

Peak is very sensitive to volatility of underlying asset.

And can be much higher than Average.

97.5% = Average + 2 x Standard Deviation

- **Typical "square-root" profile**:
  $\text{Peak}(t, 97.5\%) \sim \sigma\sqrt{t}$



Foundations: Exposure

Exposure

- Loss on default assuming zero recovery
- $E(t) = \max(V(t), 0) = V(t)^+$
  - Single trade with no collateral
  - $V(t)$ is the risk-free value (mark-to-market) of the trade at time t

Why the max?

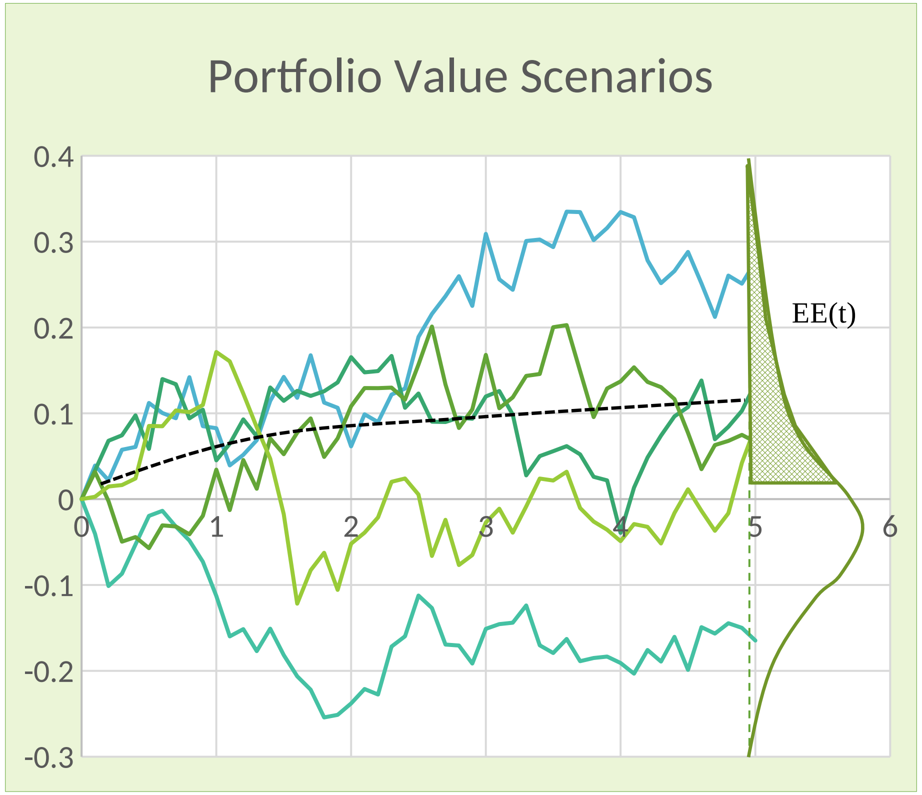
Current Exposure: $E(t_0) = V_0^+$

- Amount we would lose if default happened today

Expected Exposure Profile

- Curve of expected exposure for all future t as of today t_0:
  - $\mathrm{EE}(t) = \mathbb{E}_0^*[E(t)|t = \tau] = \mathbb{E}_0^*[V(t)^+|t = \tau]$
- **Conditional on counterparty defaulting at time $t = \tau$**
- If exposure is independent of default:
  - $\mathrm{EE}(t) = \mathbb{E}_0^*[V(t)^+]$

What is the correct measure for the expectation?



Foundations: Exposure

$$E(t) = \max(V(t), 0) = V(t)^+$$

Trade Name	Credit End Date	Product Type	Recalc MTM
fxfwdOTM_Rec 1,000,000 USD / Pay 95,000,000 INR (MP20240425000155463)	04/22/2025	FX FWD	-116,881.11

Sample Date	...	NET FWD MTM
04/19/2024	.	**-116,881.11**
04/20/2024	.	**-116,913.89**
04/21/2024	.	**-116,932.07**
04/22/2024	.	**-116,949.71**
04/23/2024	.	**-116,967.66**
04/24/2024	.	**-116,986.24**
04/25/2024	.	**-117,004.12**
05/01/2024	.	**-117,112.54**
05/08/2024	.	**-117,239.01**
05/15/2024	.	**-117,367.07**
05/22/2024	.	**-117,488.96**
06/05/2024	.	**-117,745.88**
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09/11/2024	.	**-119,530.73**
10/09/2024	.	**-120,017.70**
11/06/2024	.	**-120,546.33**
12/04/2024	.	**-121,046.05**
01/01/2025	.	**-121,535.59**

Summary **Base Exposure Detail** 0CP Exposure Detail FV

Total Standalone(Add)

Portfolio Exposure Profile(USD)

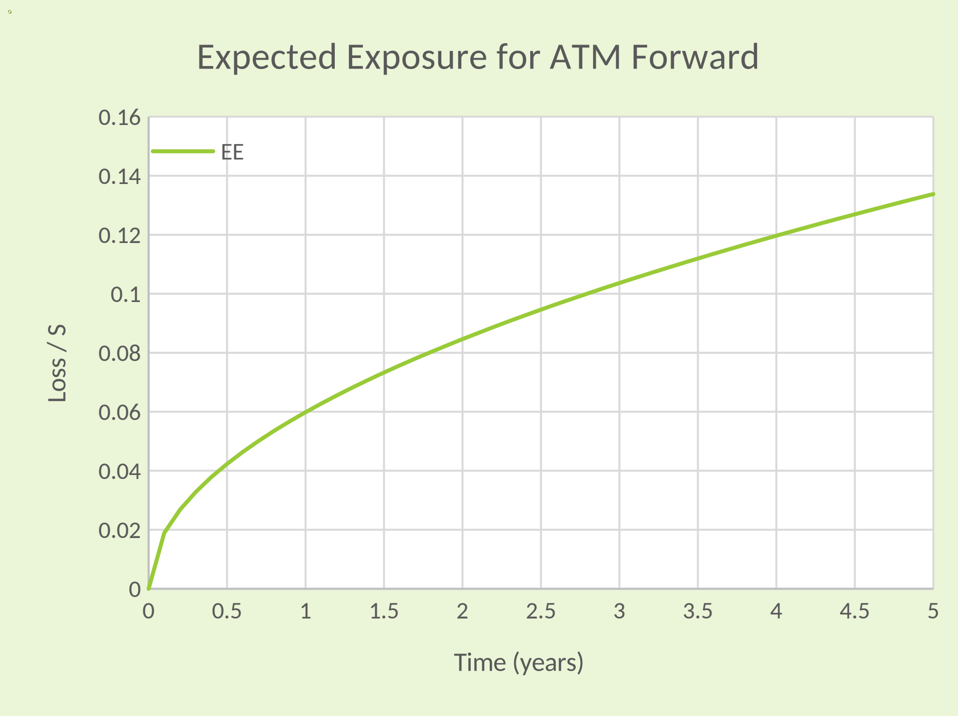
Sample Date	Peak	Average
04/19/2024	0.00	0.00
04/20/2024	0.00	0.00
04/21/2024	0.00	0.00
04/22/2024	0.00	0.00
04/23/2024	0.00	0.00
04/24/2024	0.00	0.00
04/25/2024	0.00	0.00
05/01/2024	0.00	0.00
05/08/2024	0.00	0.00
05/15/2024	0.00	0.00
05/22/2024	0.00	0.00
06/05/2024	0.00	0.00
06/19/2024	0.00	0.00
07/03/2024	0.00	0.00
07/17/2024	0.00	0.00
07/31/2024	0.00	0.00
08/14/2024	0.00	0.00
09/11/2024	0.00	0.00
10/09/2024	0.00	0.00
11/06/2024	0.00	0.00

Foundations: Exposure: Forward

Exposure profile of a forward contract

- Expected Exposure: $\mathrm{EE}(t) = \mathbb{E}_0[V(t)^+]$
- Forward on stock S_t with maturity T:
  - $V(t) = \mathbb{E}_t[S_T - k] = S_t - k$
    - *Assume zero interest rates and:* $dS_t = \sigma S_t dW_t$
- $\mathrm{EE}(t) = \mathbb{E}_0[V(t)^+] = \mathbb{E}_0[(S_t - k)^+] = \mathrm{BS}_{S_0,k}(\sigma, t)$
- **Expected exposure of forward is value of option!**
  - Exposure increases up to maturity of the trade
    - Uncertainty in value greater in future
  - $\mathrm{BS}_{S_0,k}(\sigma, t) \approx \frac{1}{2\sqrt{2\pi}} S_0 \sigma \sqrt{t}$
    - *(for small σ and t and $S_0 = k$)*
  - **Typical "square-root" profile**: $\mathrm{EE}(t) \sim \sqrt{t}$

Exercise: Derive the expected exposure for an option



$\sigma = 30\%$

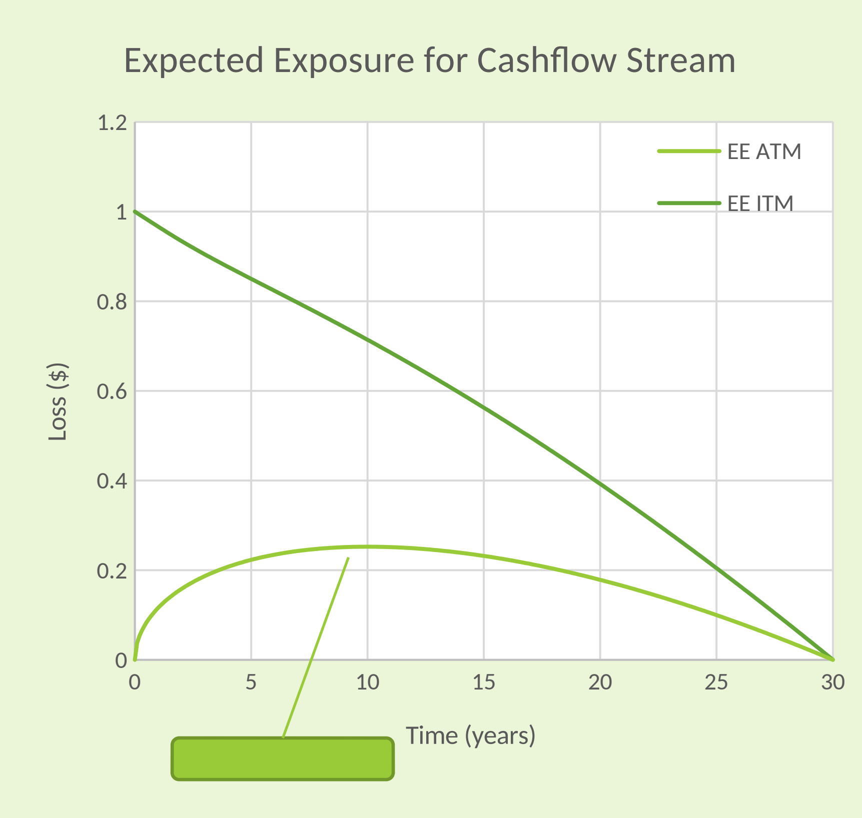
Stream of cashflows
Profile $\sim (T-t)\sqrt{t}$

Foundations: Exposure: Cashflow Stream

Exposure profile of a stream of cashflows (swap)

- Typical for a large portfolio
- Value of continuous stream of cashflows x_u with maturity T:
  - $V(t) = \mathbb{E}_t\left[\int_t^T x_u du\right] = x_t(T-t)$ and $V_0 = x_0 T$
  - *Assume:* $r = 0$ *and* $dx_u = \sigma^x dW_u$ *(*x_t *is a martingale)*
- $\mathrm{EE}(t) = \mathbb{E}_0\left[(T-t)x_t^+\right] = (T-t)\mathbb{E}_0\left[(x_0 + \sigma W_t)^+\right]$
  - ➢ $\mathrm{EE}(t) = (T-t)\sigma^x\sqrt{t}\left(\tilde{x}_0\Phi(\tilde{x}_0) + \phi(\tilde{x}_0)\right)$ with $\tilde{x}_0 = \frac{x_0}{\sigma^x\sqrt{t}}$

At the Money:	In the Money:	Out of the Money:
• Exposure profile has typical hump shape	• Linearly decreasing profile starting at today's value	• No exposure



$\sigma^x = 1\%$

Stream of cashflows Profile$\sim(T-t)\sqrt{t}$

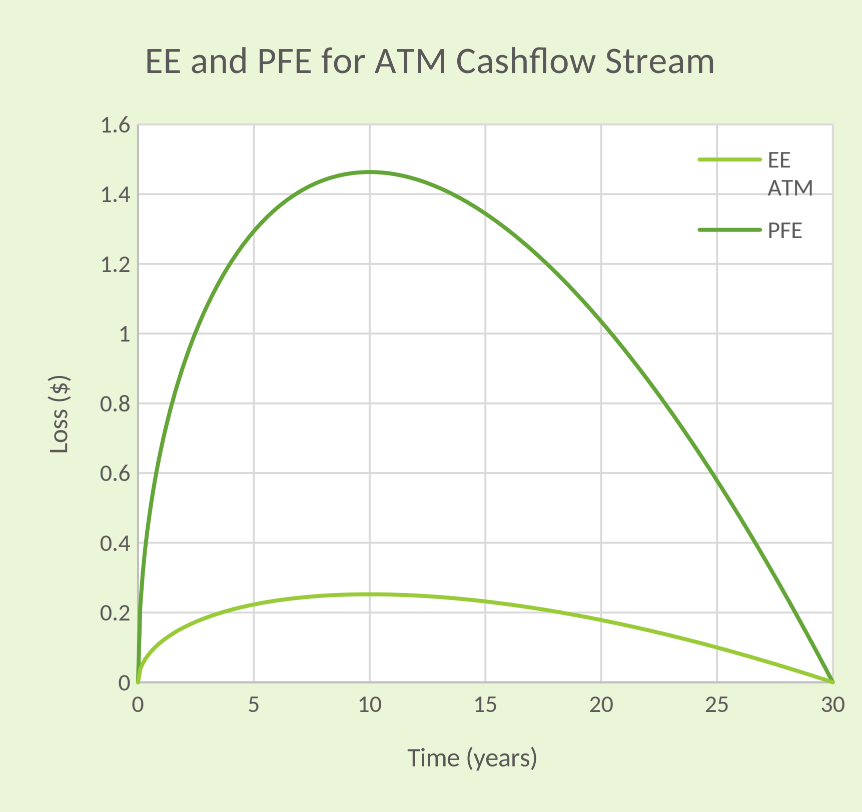
Foundation:
Other Exposure Metrics

Expected Exposure

- Fundamental input to XVA calculations
- Also used in Basel Default RWA definition
  - Exposure at Default (EAD) is function of EE in first year

Potential Future Exposure (PFE)

- "Worst case loss" defined as quantile in exposure distribution
- Used to set credit risk limits
- PFE at quantile q and time t defined by:
  - $\mathrm{Prob}[E(t) < \mathrm{PFE_q}(t)] = q$
  - Typically, $q = 99\%$
- If $V(t)$ is Gaussian with zero mean, then
  - ➢ $\mathrm{PFE}_{99\%}(t) \approx 5.8 \cdot \mathrm{EE}(t)$



$\sigma^x = 1\%$

Foundations:
Risk Mitigation: Netting

What is the exposure of portfolio with multiple trades?

- Depends on "**netting agreement**" (**ISDA agreement**)

Netting is contractual agreement allowing the value of multiple trades with one counterparty to be added prior to making a default claim

  - Exposure without netting: $\mathrm{EE}(t) = \sum_i \mathbb{E}_0^*[V_i(t)^+]$
  - Exposure with netting: $\mathrm{EE}(t) = \mathbb{E}_0^*[(\sum_i V_i(t))^+]$

- Netting is always risk reducing
  - Can net across all product types with single counterparty (bi-lateral netting)
- Benefit scales with portfolio size
  - Assume N trades with independent, Gaussian distributions for the values V_i
  - Expected exposure without netting scales like N; with netting scales like $\sqrt{N}$
- **To benefit, we need to manage Counterparty Risk centrally**
  - Exposure needs to be calculated at the counterparty portfolio level (netting set)

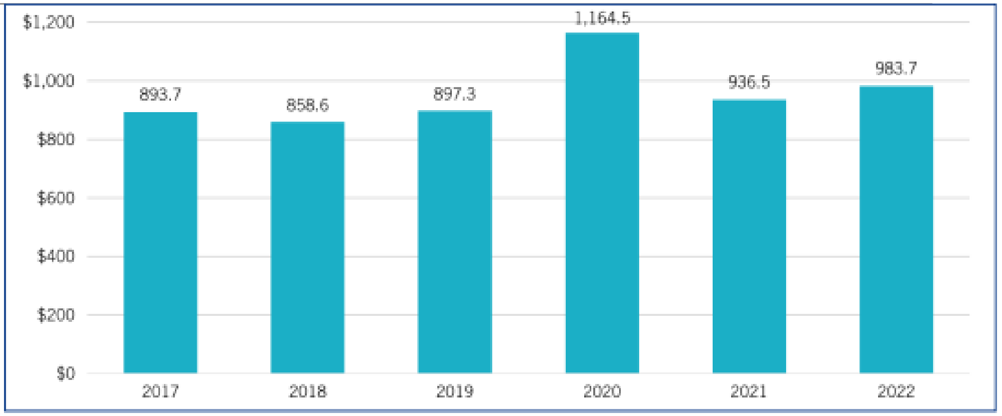
$$\underbrace{\left(V_1(t) + V_2(t)\right)^+}_{\text{Exposure with netting}} < \underbrace{V_1(t)^+ + V_2(t)^+}_{\text{Exposure without netting}}$$

	Trade Values		Exposure	
			No Netting	Netting
P1	20	5	25	25
P2	20	-5	20	15
P3	-20	5	5	0

Foundations:
Risk Mitigation: Collateral

Collateral is cash or assets posted to offset default risk

- The ***Credit Support Annex*** **(CSA)** defines the collateral terms
  - Can be cash or high quality assets (e.g. government securities)
  - Collateral amounts are updated on a regular basis
    - Typically have a daily collateral frequency
    - Posting amount depends on the amount and direction of the risk
  - Collateral can be bi-lateral or uni-lateral
  - Collateral can be re-used: It can be "re-hypothecated"
- Collateral directly reduces the exposure:
  - $E(t) = \left(\mathrm{V}(t) - \mathrm{C}(t)\right)^{+}$
  - where $C(t)$ is the collateral held at time t
- Two main types of collateral:
  - Variation Margin (VM)
  - Initial Margin (IM)
- **However**, large collateral calls can cause distress
  - Can transmit default risk



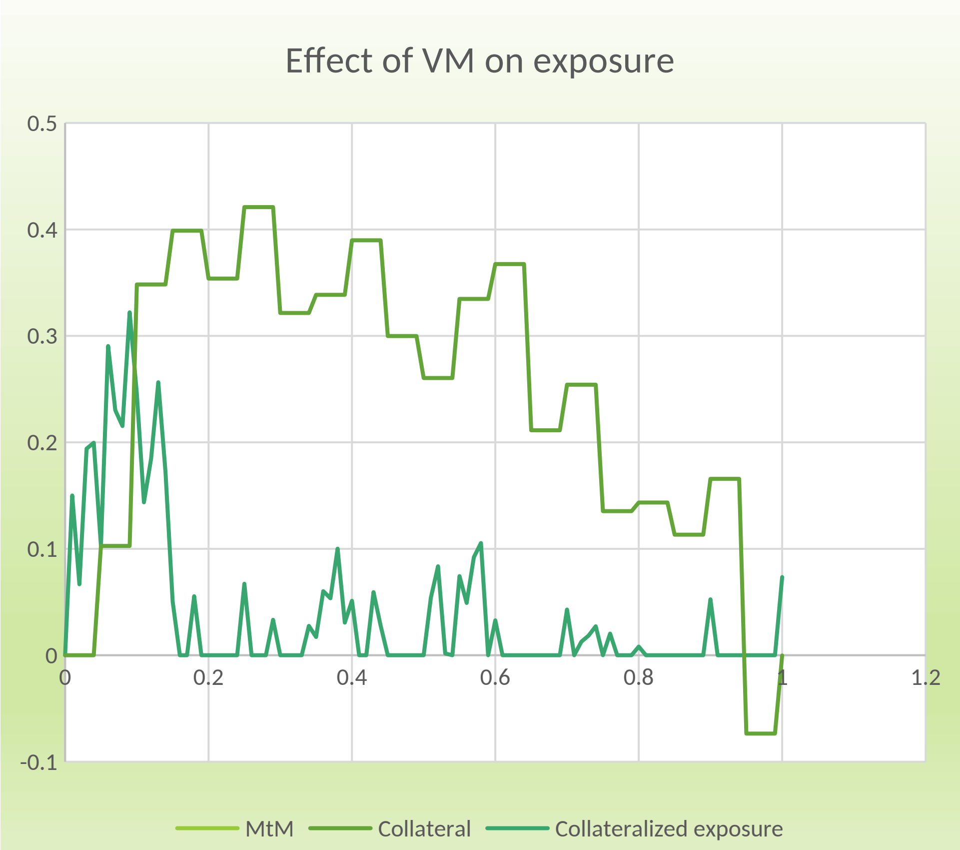
Total VM received by Phase-one Firms (US$ billions), *ISDA Margin Survey 2022*

Regulatory IM and IA received by Phase-one Firms (US$ billions)

Foundations:
Risk Mitigation: Variation Margin

Variation Margin (VM)

- Collateral to cover mark-to-market (MtM) moves.
- Typically posted daily and equal to the MtM on that day
- If a counterparty defaults, there will be a delay between the last collateral payment and the default settlement
  - This is the "margin period of risk (MPoR)" Δ
  - Also referred to as the "cure period"
- Exposure given by:
  - $E(t) = \left(\mathrm{V}(t) - \mathrm{C}(t)\right)^{+} = \left(\mathrm{V(t)} - V(\mathrm{t} - \Delta)\right)^{+}$
- Retain "cure period risk"
  - Sensitive to MtM forward volatility over cure period Δ
- Can become important in stress scenarios
- Variation margin is standard between financial counterparties



Foundations:
Variation Margin Exposure

$$E(t) = \max(V(t), 0) = V(t)^+$$

$$E(t) = \big(\mathrm{V}(t) - \mathrm{C}(t)\big)^+ = \big(\mathrm{V}(\mathrm{t}) - V(\mathrm{t} - \Delta)\big)^+$$

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Total Standalone(Add)

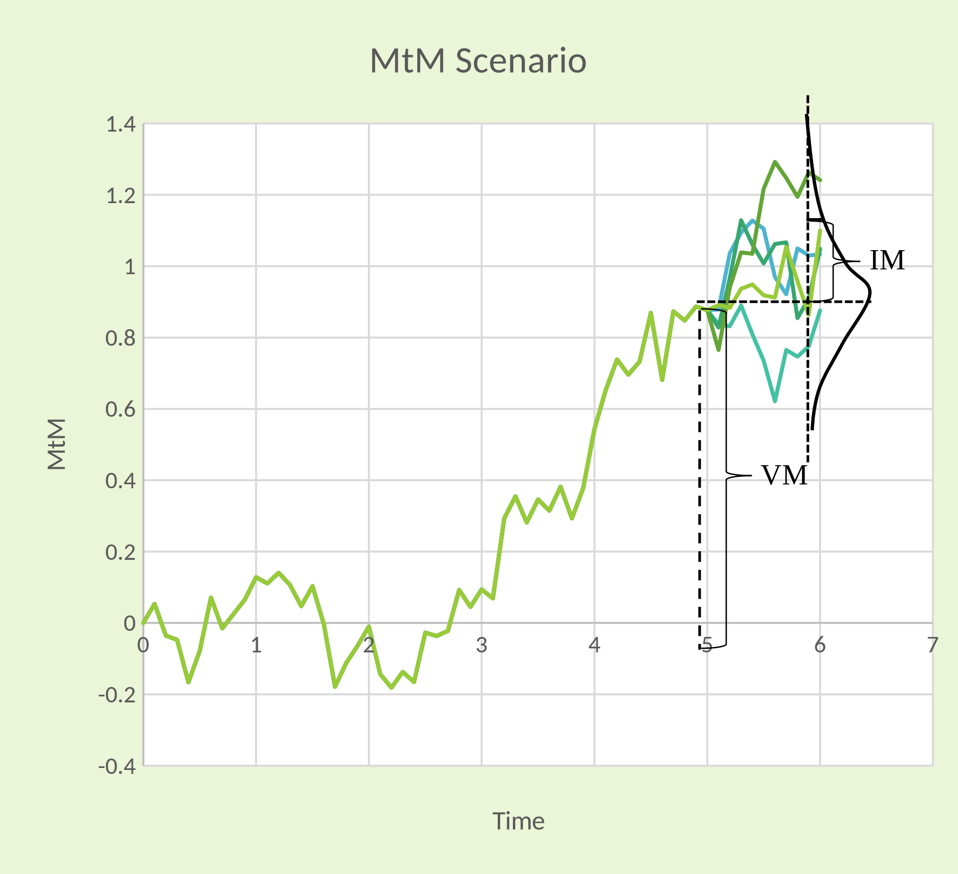
Portfolio Exposure Profile(USD)

Sample Date	Peak	Average
04/19/2024	0.00	0.00
04/20/2024	2,714.71	548.90
04/21/2024	5,241.89	1,047.84
04/22/2024	6,463.15	1,320.42
04/23/2024	7,584.68	1,557.90
04/24/2024	8,940.63	1,802.32
04/25/2024	9,868.36	1,995.60
05/01/2024	13,634.64	2,723.20
05/08/2024	13,480.53	2,665.25
05/15/2024	12,722.29	2,512.47
05/22/2024	13,131.46	2,579.33
06/05/2024	14,639.63	2,869.21
06/19/2024	14,576.71	2,874.44
07/03/2024	14,529.06	2,893.26
07/17/2024	14,639.40	2,934.32
07/31/2024	15,676.56	3,165.20
08/14/2024	15,051.36	3,081.98
09/11/2024	15,139.23	3,129.10

Foundations:
Risk Mitigation: Initial Margin

Initial Margin (IM)

- Additional collateral to cover cure period risk
- Can be static or dynamic
- $E(t) = \left(\mathrm{V}(t) - \mathrm{C}(t)\right)^{+} = (\mathrm{V}(\mathrm{t}) - V(\mathrm{t} - \Delta) - \mathrm{IM}(t))^{+}$
- IM is usually calculated using risk sensitive models
  - E.g. quantile in distribution of $\Delta V(t)$
- IM is mandated by regulators for large financial firms
  - Standard Industry Margin Model (SIMM)
- IM is charged by CCPs
- Residual exposure with IM is difficult to model
  - Required to achieve regulatory capital benefit



Foundations: Recovery and Loss

Exposure is the amount that is owed upon counterparty default

- This is the basis of **default claim** made to the counterparty
  - Also referred to as "**close-out amount**"
- Remaining assets of counterparty will be distributed to creditors
- Derivatives claims are treated like loans (not bonds)
  - They are "**pari passu**" with loans
- Actual loss on default is a fraction of the claim
- The amount recovered is given by the **recovery rate** R^C
  - **Loss given default**: $\text{LGD} = (1 - R^C)$
  - Typically: $R^C \approx 40\%$
- Loss is given by: $\boldsymbol{L(t) = \left(1 - R^C\right)E(t)}$
- *What is the value or cost of the expected loss?*

Valuation: Overview

- **CVA**
  - What is the value of the expected loss?
    - How do we adjust derivative valuation?
- **Examples**
  - How do we calculate CVA?
    - Approximations and intuition
- **WWR**
  - What is Wrong-Way-Risk?
    - How does it impact exposure?

Valuation: CVA Definition

CVA: **Credit Valuation Adjustment** = Expected loss due to counterparty default

- CVA captures the cost or value associated with a potential default loss
  - Cost of hedging the risk
- CVA is an add-on to the risk-free value
- Define CVA as the value of a derivative with payoff equal to loss on default

$$\boxed{\textbf{CVA}=\mathbb{E}^{\boldsymbol{Q}}\left[\boldsymbol{D_0}(\boldsymbol{\tau})\left(\mathbf{1}-\boldsymbol{R^C}\right)\boldsymbol{E}(\boldsymbol{\tau})\mathbb{I}_{\boldsymbol{\tau}<\boldsymbol{T}}\right]}$$

  - $D_t(T) = e^{-\int_t^T r_s ds}$ is the stochastic discount factor with riskless rate r_t
  - τ is the default time of the counterparty and T is the maturity of the derivative
  - $\mathbb{I}$ is the indicator function
  - $\mathbb{E}^Q$ is the expectation in the risk neutral measure
- This is “unilateral” CVA as our own default is not considered

Valuation:
CVA as Valuation Adjustment

CVA is an adjustment to the risk-free trade value $\boldsymbol{V_{RF}}$

- $\boldsymbol{V_{tot} = V_{RF} - CVA}$
- **Allows separation of trading desk and CVA desk**

$$V_{RF}(t) \equiv \mathbb{E}_t^Q[D_t(T)\Phi(T)]$$

Derivation for derivative with payoff $\Phi(T)$ at maturity T:

$$V_{RF}(\tau)^- \equiv -\min(V_{RF}(\tau), 0) \\ = -(V_{RF}(\tau) - V_{RF}(\tau)^+)$$

$$V_{tot} = \underbrace{\mathbb{E}_0^Q[D_0(T)\Phi(T)\mathbb{I}_{\tau>T}]}_{\text{no default}} + \underbrace{\mathbb{E}_0^Q[D_0(\tau)(R^C V_{RF}(\tau)^+ - V_{RF}(\tau)^-)\mathbb{I}_{\tau<T}]}_{\text{default}}$$

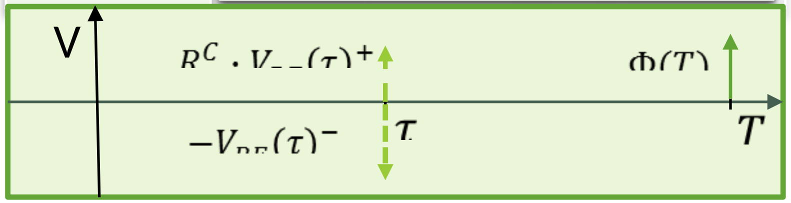
$$\begin{aligned} V_{tot} &= \mathbb{E}_0^Q[D_0(T)\Phi(T)\mathbb{I}_{\tau>T}] + \mathbb{E}_0^Q[D_0(\tau)V_{RF}(\tau)\mathbb{I}_{\tau<T}] - \mathbb{E}_0^Q[D_0(\tau)(1-R^C)V_{RF}(\tau)^+\mathbb{I}_{\tau<T}] \\ &= \mathbb{E}_0^Q[D_0(T)\Phi(T)] - \mathbb{E}_0^Q[D_0(\tau)(1-R^C)E(\tau)\mathbb{I}_{\tau<T}] \\ &= V_{RF} - CVA \end{aligned}$$

$$\begin{aligned} \mathbb{E}_0^Q[D_0(\tau)V_{RF}(\tau)\mathbb{I}_{\tau<T}] &= \mathbb{E}_0^Q\left[D_0(\tau)\mathbb{E}_\tau^Q[D_\tau(T)\Phi(T)]\mathbb{I}_{\tau<T}\right] \\ &= \mathbb{E}_0^Q\left[\mathbb{E}_\tau^Q[D_0(\tau)D_\tau(T)\Phi(T)\mathbb{I}_{\tau<T}]\right] \\ &= \mathbb{E}_0^Q\left[D_0(T)\Phi(T)\mathbb{I}_{\tau<T}\right] \end{aligned}$$

Default

No Default

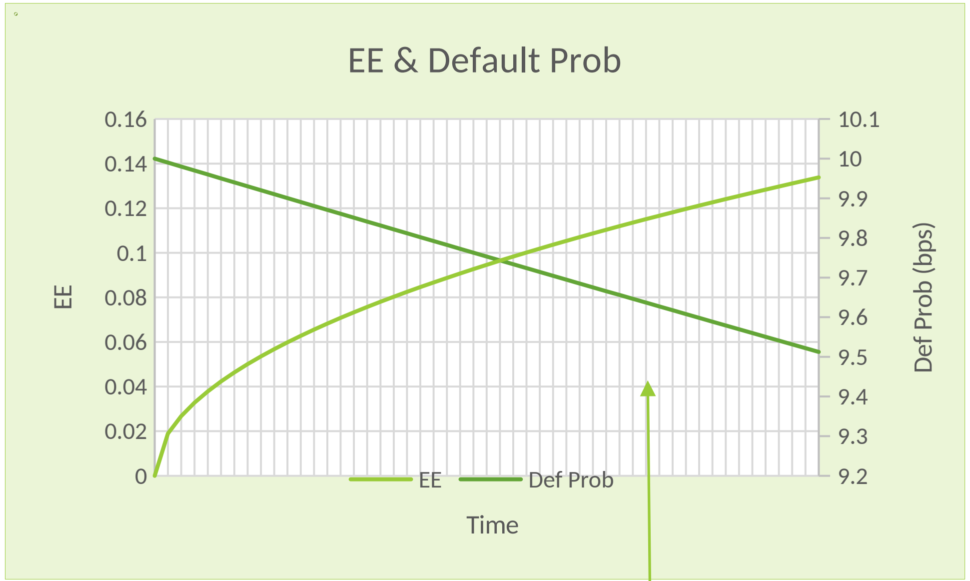
	Payoff $V>0$	Payoff $V<0$
Default	$R^C \cdot V_{RF}(\tau)$	$V_{RF}(\tau)$
No Default	$V(T) = \Phi(T)$	



CCR Basics
From CCR to CVA

CVA stands for **Credit Valuation Adjustment**

- Defined as the value of the expected loss
- Loss on default of counterparty C:
  - $L(t) = (1 - R^C)E(t)$
  - Where R^C is the **recovery rate**



$\text{Prob}[t < \tau^C < t + dt] = \lambda^C(t)Q^C(t)$

Uni-lateral CVA

- Assume we are riskless:

Discount factor: $D_t(T) = e^{-\int_t^T r_s^* ds}$

$$\text{UCVA}=\mathbb{E}_0^Q\left[D_0(\tau)(1 - R^C)E(\tau)\mathbb{I}_{\tau<T}\right]$$

Default time: τ

- Can be written as time integral:

Zero coupon bond price: $Z(t) = \mathbb{E}^Q[D_0(t)]$

$$\text{UCVA} = (1 - R^C)\int_0^T \lambda^C(t)Q^C(t)Z(t)\text{EE}^+(t)dt$$

Default intensity: λ^C

Survival probability: $Q^C(t) = e^{-\int_0^t \lambda^C(s)ds}$

Valuation:
CVA as a stream of CDS

CVA is a time integral of the expected exposure

- CVA=$\mathbb{E}_0^Q[D_0(\tau)(1-R^C)E(\tau)\mathbb{I}_{\tau<T}]$
- CVA=$(1-R^C)\int_0^T \mathbb{E}_0^Q[D_0(t)E(t)|\tau=t]P(\tau=t)dt$
- CVA=$(1-R^C)\int_0^T Z(t)\mathrm{EE}(t)P^C(\tau=t)dt$

$$\text{CVA}=(1-R^C)\int_0^T \lambda^C(t)Q^C(t)Z(t)\mathrm{EE}(t)dt$$

$$\text{CVA}=(1-R^C)\int_0^T \lambda^C(t)e^{-\int_0^t(r(s)+\lambda^C(s))ds}\mathrm{EE}(t)dt$$

- CVA can be valued as the contingent leg of a CDS with time varying notional
- CVA can be replicated with a term structure of CDS

Where:

- $Z(t)$ is the zero-coupon bond price
  - $Z(t)=\mathbb{E}^Q[D_0(t)]=e^{-\int_0^t r(s)ds}$
  - Here $r(s)$ is the forward rate
- EE(t) is the calculated in the “t-forward” measure
  - $\mathrm{EE}(t)=\mathbb{E}^Q\left[\frac{D_0(t)}{Z(t)}E(t)|\tau=t\right]=\mathbb{E}^F[E(t)|\tau=t]$
- $P(\tau=t)dt$ is the default prob in interval *(t, t+dt)*
  - $P^C(\tau=t)dt=\lambda^C(t)Q^C(t)dt=\lambda^C(t)e^{-\int_0^t\lambda^C(s)ds}dt$
  - $Q^C(t)=e^{-\int_0^t\lambda^C(s)ds}$ is the counterparty survival probability
  - $\lambda^C(t)=-\frac{d}{dt}\ln Q(t)$ is the forward default rate of the counterparty

Valuation
CVA Hacks

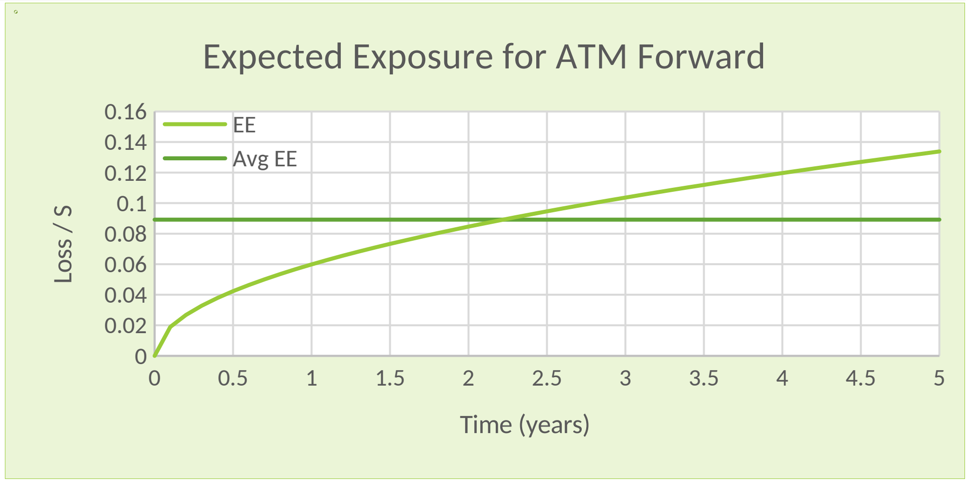
To gain intuition it is useful to simplify CVA formula

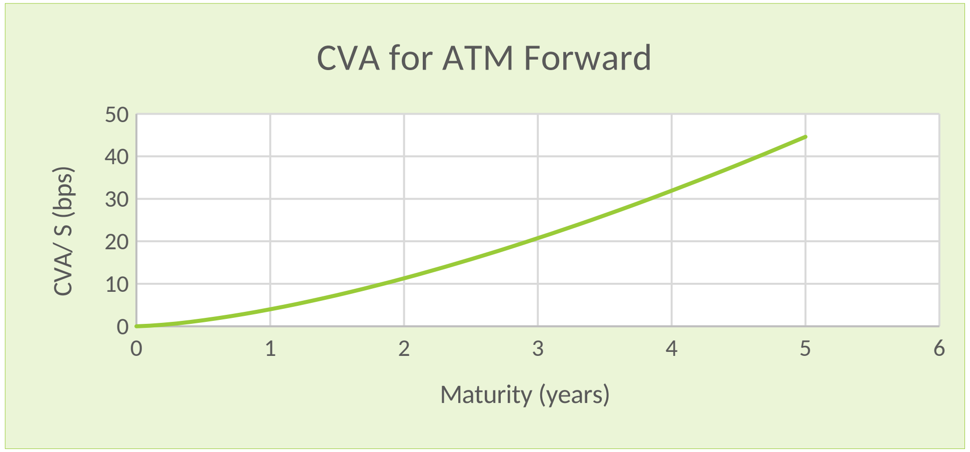
- CVA$=(1-R^C)\int_0^T \lambda^C(t) e^{-\int_0^t (r(s)+\lambda^C(s))ds} \mathrm{EE}(t)dt$
  - Assume λ and r are small, i.e. discount and survival factors are close to 1
  - Assume $\lambda^C(t) = \lambda_0^C$ is constant
  - Counterparty credit spread $s^c \approx (1-R^C)\lambda^C$
- $\mathrm{CVA} = s^c \cdot \overline{\mathrm{EE}} \cdot T$ where
- $\overline{\mathrm{EE}} \equiv \frac{1}{T}\int_0^T \mathrm{EE}(t)dt$ is the average expected exposure

Valuation:
CVA Examples: ATM Forward

$$\mathbf{CVA} = \boldsymbol{s^c} \cdot \overline{\mathbf{EE}} \cdot \boldsymbol{T}$$

- Expected Exposure:
  - $\mathrm{EE}\,(t) = BS_{S_0,k}(\sigma, t) \approx \frac{1}{2\sqrt{2\pi}}\sigma\sqrt{t}$
  - *Assume* $S_0 = k = 1$
- $\overline{\mathrm{EE}} = \frac{1}{T}\int_0^T \frac{1}{2\sqrt{2\pi}}\sigma\sqrt{t}dt = \frac{1}{3\sqrt{2\pi}}\sigma\sqrt{T} = \frac{2}{3}\mathrm{EE}_{\max}$
- $\mathrm{CVA} = \frac{1}{3\sqrt{2\pi}} s^c \cdot \sigma \cdot T^{\frac{3}{2}}$
- For typical values: $s^c = 100\text{bps}$, $\sigma = 30\%$, T=5y
  - $\mathrm{CVA} \approx 45\text{bps}$





Valuation:
CVA Examples: Cashflow Stream

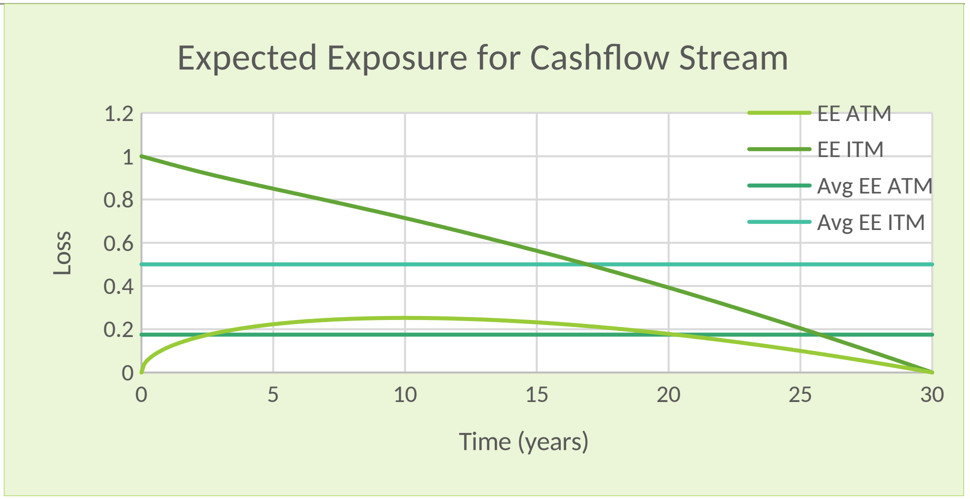
At the Money (ATM)

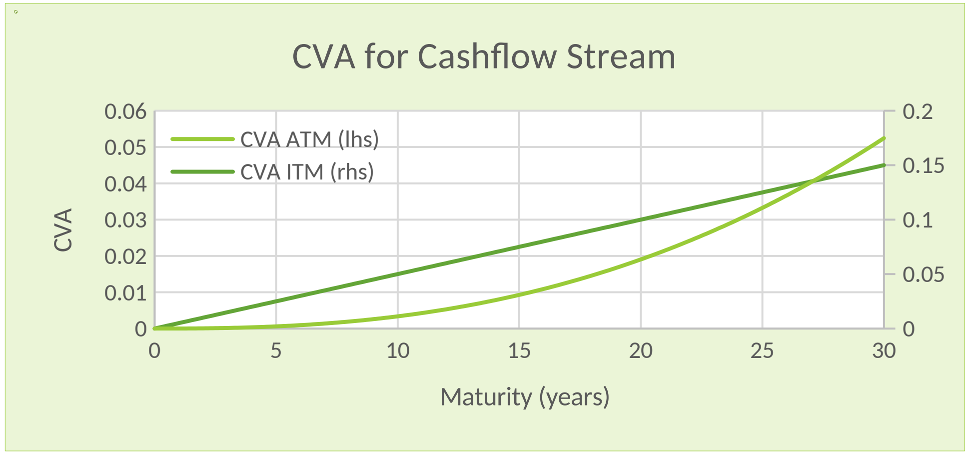
- $\mathrm{EE}\ (t) = \frac{1}{\sqrt{2\pi}} \sigma \sqrt{t}(T - t)$
- $\overline{\mathrm{EE}} = \frac{1}{T}\int_0^T \mathrm{EE}(t)dt = \frac{4}{15\sqrt{2\pi}}\sigma^x T^{\frac{3}{2}} = \frac{2\sqrt{3}}{5}\mathrm{EE}_{\max}$
- $\mathrm{CVA} = s^c \cdot \overline{\mathrm{EE}} \cdot T = \frac{4}{15\sqrt{2\pi}} s^c \cdot \sigma^x \sqrt{T} \cdot T^2$
- $\mathrm{CVA} \approx 5.2\%$ or 17bps running

In the Money (ITM)

- $\mathrm{EE}\ (t) = V_0\left(1 - \frac{t}{T}\right)$
- $\overline{\mathrm{EE}} = \frac{1}{2} V_0$
- $\mathrm{CVA} = \frac{1}{2} s^c \cdot V_o \cdot T$
- $\mathrm{CVA}/V_0 \approx 15\%$

Typical values:
$s^c = 100\text{bps}$,
$\sigma^x = 1\%$,
T=30y





Modelling: Overview

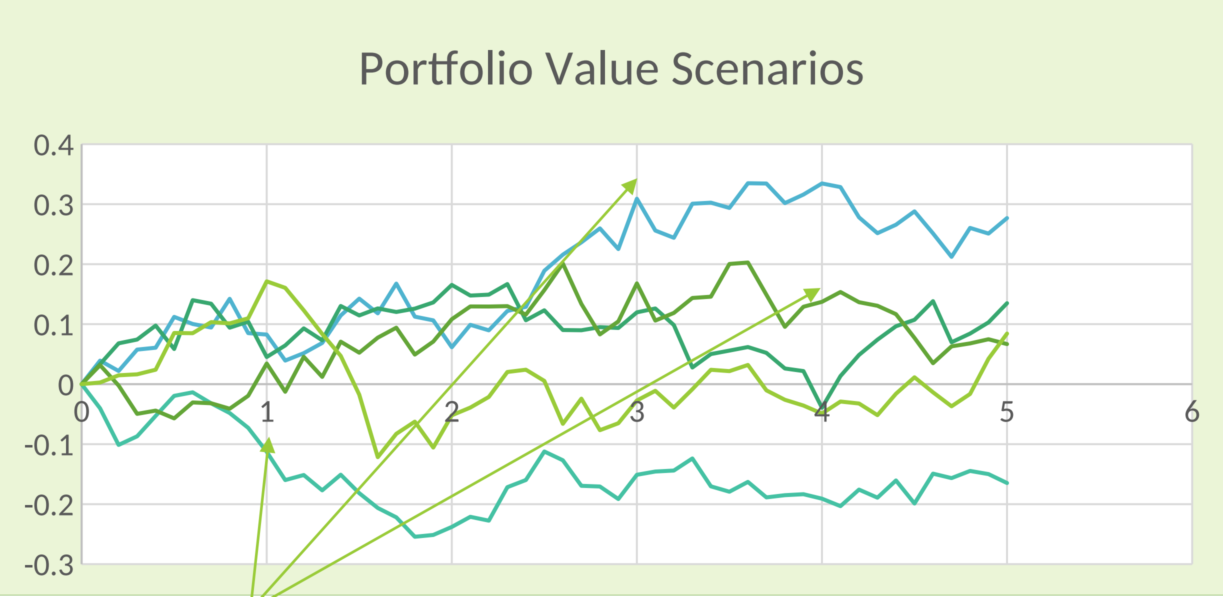
CVA is typically calculated centrally and at a counterparty portfolio level

Simulation
- Generate market scenarios using Monte Carlo simulation

Pricing
- In each scenario and a selection of future dates value all trades for all portfolios

Aggregation
- For each path and date compute exposure incorporating collateral and netting

Valuation
- Use EE profile, counterparty spreads and recoveries to calculate derived metrics
- CVA, other XVA



$$V_{tot}(t_i;\omega_j) = \sum_k V_k(t_i;\omega_j)$$

$$E(t_i,\omega_j) = \left(V_{tot}(t_i;\omega_j) - C(t_i;\omega_j)\right)^+$$

$$\mathrm{EE}(t_i) = \frac{1}{N_\omega}\sum_{j=1}^{N_\omega} E(t_i;\omega_j)$$

$$\mathrm{CVA} = (1 - R^C)\sum_{i=1}^{N_T} \lambda^C(t_i)Q^C(t_i)Z(t_i)\mathrm{EE}(t_i)\Delta t_i$$

Modelling: Spreads and Recoveries

Default Probabilities

- Use market implied counterparty default probabilities for valuation
- Bootstrap probabilities from CDS spreads if available
  - $\text{Spread} = (1 - R)\lambda$
- Few counterparties will have liquid CDS
- Use "proxy" spreads
  - Related counterparties
  - Indices
- Or interpolate spreads using:
  - Rating, industry & sectorial classifications
  - Bond data. Equity?
  - Use regression or ML techniques
- Risk management will rely on indices or proxy CDS (macro hedging)

Recovery rates

- Market credit spreads typically imply a fixed recovery of around 40%.
- Market recovery rate should be used for bootstrapping default probabilities
- Market recovery rates refer to bond recoveries
- $\text{CVA} = (1 - R^C) \int_0^T Z(t) EE(t) \lambda^C(t) e^{-\int_0^t \lambda^C(s) ds} dt$
- Recovery rate for derivative may be different
  - Derivatives are bi-lateral instruments and typically *pari passu* with loans

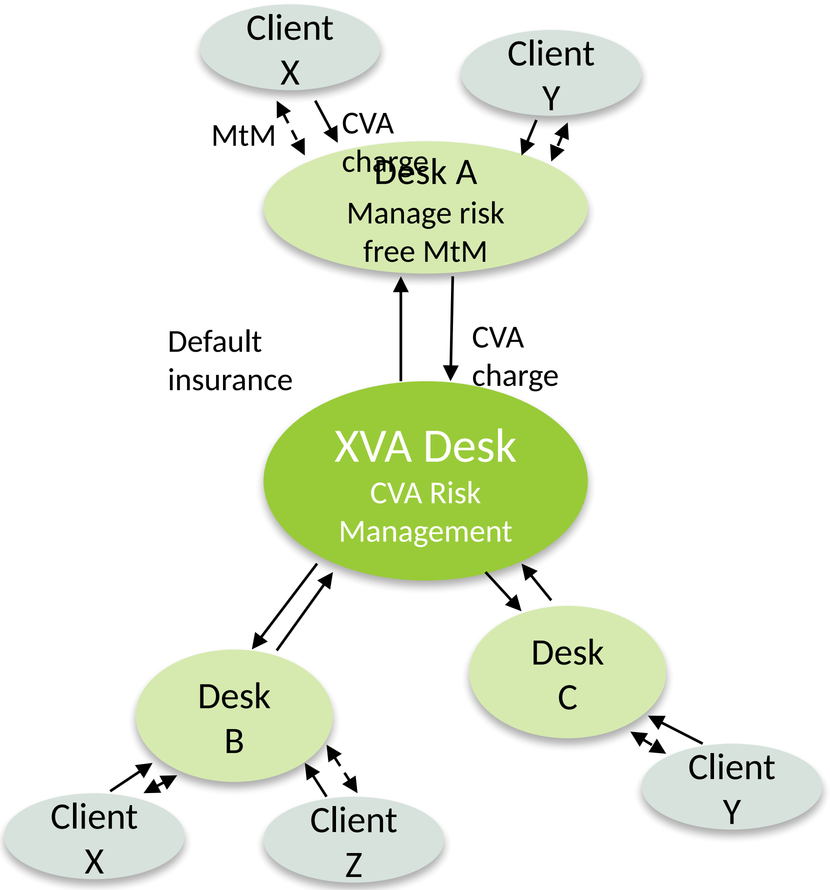
CVA Practice:
Charging and Allocation

To capture portfolio level collateral terms and benefit from netting:

- Calculate CVA at counterparty level
- Calculate and risk manage CVA centrally
- CVA desk insures defaults in return for the CVA charge

However, client charges are at trade level and managed by business units

- Require trade allocation for CVA
- Incremental (marginal) exposure for trade A:
  - $EE_A(t) = \mathbb{E}[(V_A(t) + V_{tot})^+] - \mathbb{E}[V_{tot}^+]$
  - Portfolio value $V_{tot} = \sum_i V(t)$
- Depends on order of trades
- Requires optimization of compute & memory overhead
  - ➢ Precompute V_{tot} for each counterparty



CVA Practice: Risk Management

A CVA trading desk will calculate numerous metrics for risk management

- CVA Greeks/Sensitivities: $\frac{\partial CVA}{dX_i}$
- P&L Explain and Predict: $CVA(t) \approx CVA(t - \Delta t) + \sum_i \frac{\partial CVA}{dX_i} \Delta X_i$
- CVA VaR and Stress: quantile in CVA distribution or CVA in specific stress scenario

Spread Risk

- CVA is linear in spreads. Risk is efficient to compute
- Hedge with CDS either single name or index (macro hedging)
- Can only hedge idiosyncratic default risk when counterparty has liquid CDS instruments
- Otherwise, can only hedge spread MtM risk

Market Risk

- Large number of risk factors. Each risk factor sensitivity requires full re-valuation
- Equivalent to exotic derivative hedging
- Very expensive if not optimized e.g. using analytic sensitivities (AAD)
- Focus on key risk drivers, e.g. IR, FX. Hedge using swaps and options

CVA Practice: Performance

- Credit Exposure calculations are highly compute intensive.
- E.g. 1M trades, 10k paths, 100 time points -> 1Trillion evaluations
  - Assume 1ms per trade -> require 275k CPU hours
  - Require cutting edge compute architecture:

Code Optimization	Hardware
• Fast pricers • Monte Carlo variance reduction • Analytical Adjoint Differentiation (AAD) • Trade compression	• Parallelization • GPUs • Cloud Computing • Large memory machines • Quantum computing…

Conclusion (CVA)

- Counterparty risk is the default risk for derivatives portfolios
- CVA is an adjustment to the risk-free trade value to account for counterparty risk
- CVA calculation is similar to pricing an exotic derivative
- Presence of netting and collateral requires portfolio level risk management
- Exposure calculations are highly compute intensive and require advanced software and hardware
- CVA risk management in particular focus in times of crisis

References

- John Gregory, *The xVA Challenge: Counterparty Risk, Funding, Collateral, Capital and Initial Margin*, 4th Edition (Wiley Finance), 2020.
- Andrew Green, *XVA: Credit, Funding and Capital Valuation Adjustments*, (The Wiley Finance Series), 2015.

FVA

Funding
What is Funding?

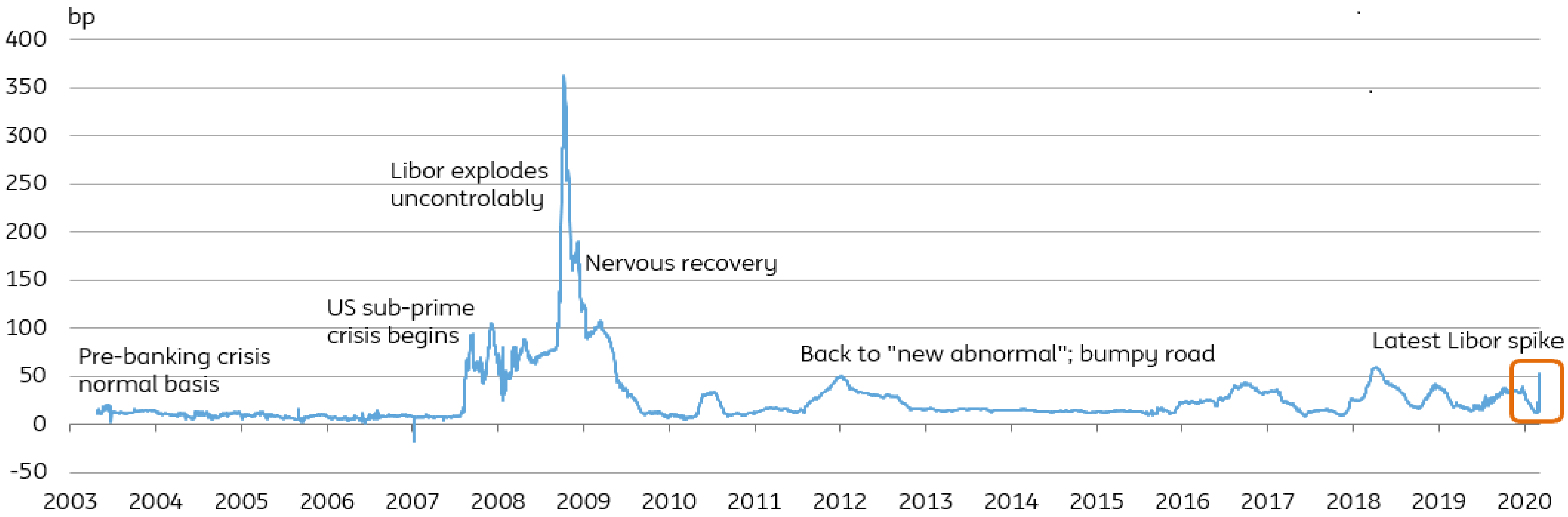
Funding is the act of providing resources (money) to finance a business

- Two types of funding: Equity and Debt
  - A company can issue shares or sell bonds to raise money
- Typical example
  - A bank makes a $100 loan to a corporate client.
  - To make the loans the bank will need to raise funds (typically via deposits – a form of debt)
- Derivatives business
  - Holding and risk managing a derivative to maturity requires ongoing and dynamic funding
  - This is represented by the "bank account" in the replicating portfolio
  - Funding for the derivatives business is provided by the **Treasury** function.
  - The Treasury charges the business a "**funding rate**" for the financing
- The funding rate is based on the rate that the firm can borrow money in the inter-bank market
- All derivatives funding is assumed to be via debt

Funding
The Funding Rate

The funding rate is linked to the default probability of the firm

- $r^F = r^* + s^F$, where $\boldsymbol{s^F} = (\mathbf{1} - \boldsymbol{R})\boldsymbol{\lambda}$ is the funding spread
- The spread is compensation for the expected loss due to default



BCVA & DVA
From Uni-lateral to Bi-lateral CVA

Uni-lateral CVA: UCVA

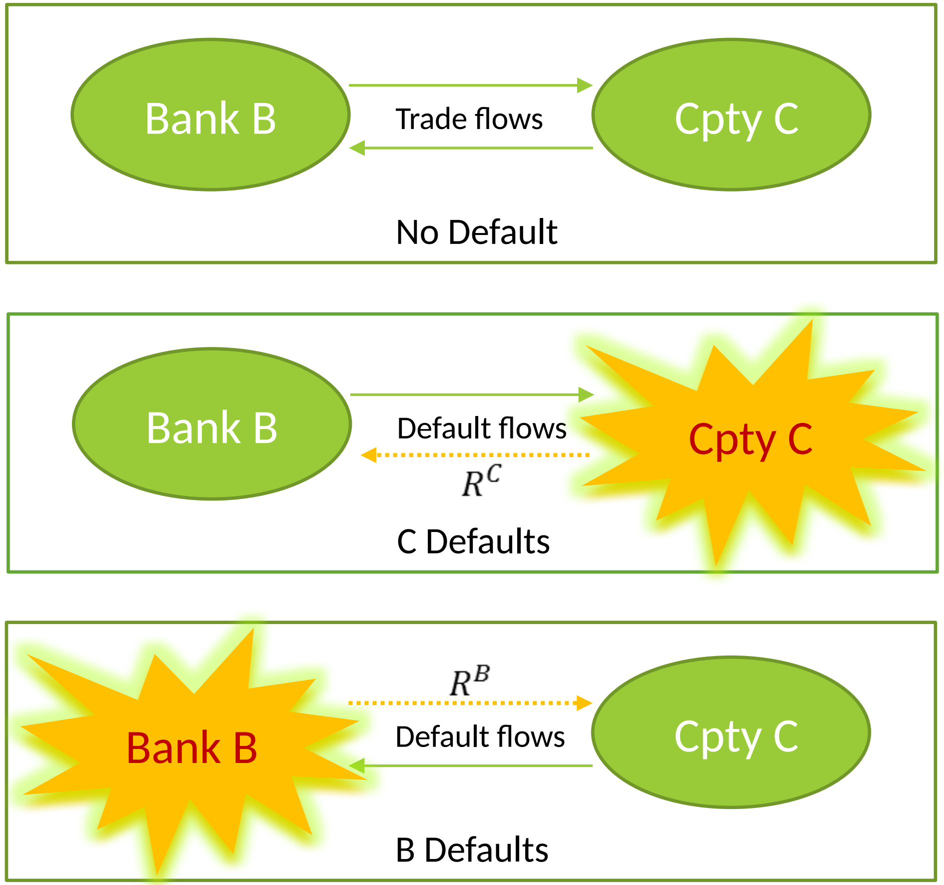
- So far we have assumed that we are “risk free”
- But we are also a counterparty!
- Uni-lateral CVA breaks valuation symmetry

What happens if we default?

- Only have counterparty risk conditional on our survival
  - Adjust UCVA by our survival probability to get bi-lateral CVA: BCVA
- In default we no longer need to pay counterparty
  - This is a benefit -> Debt Valuation Adjustment (DVA)
  - Benefit is the CVA from the counterparty perspective

What is the total value?

- Expect: $V_{tot} = V^* - \mathrm{BCVA} + \mathrm{DVA}$
- This is symmetric between counterparties



BCVA & DVA
Bi-lateral CVA derivation 1

Consider single derivative trade with payoff $\Phi(T)$ at maturity T

- Assume both we (Bank B) and Counterparty C are risky

$$V^*(t)^+ \equiv \max(V^*(t), 0)$$
$$V^*(t)^- \equiv -\min(V^*(t), 0)$$
$$V^*(t) = V^*(t)^+ - V^*(t)^-$$

B receives/pays	Time	$V^* > 0$	$V^* < 0$
I: No default	T	$\Phi(T)$	
II: C defaults first	τ^C	$R^C \cdot V^*(\tau^C)^+$	$-V^*(\tau^C)^-$
III: B defaults first	τ^B	$V^*(\tau^B)^+$	$-R^B \cdot V^*(\tau^B)^-$

$$V_{tot} = \underbrace{\mathbb{E}^Q\left[D_0(T)\Phi(T)\mathbb{I}_{\tau_B,\tau_C>T}\right]}_{\text{I: no default}} + \underbrace{\mathbb{E}^Q\left[D_0(\tau^C)(R^C V^*(\tau^C)^+ - V^*(\tau^C)^-)\mathbb{I}_{\tau^C<T}\mathbb{I}_{\tau^C<\tau^B}\right]}_{\text{II: C defaults first}}$$

$$+ \underbrace{\mathbb{E}^Q\left[D_0(\tau^B)(V^*(\tau^B)^+ - R^B V^*(\tau^B)^-)\mathbb{I}_{\tau^B<T}\mathbb{I}_{\tau^B<\tau^C}\right]}_{\text{III: B defaults first}}$$

$$\mathrm{II} = \mathbb{E}^Q\left[D_0(\tau^C)V^*(\tau^C)\mathbb{I}_{\tau^C<T}\mathbb{I}_{\tau^C<\tau^B}\right] - \underbrace{\mathbb{E}^Q\left[D_0(\tau^C)(1-R^C)V^*(\tau^C)^+\mathbb{I}_{\tau^C<T}\mathbb{I}_{\tau^C<\tau^B}\right]}_{\text{BCVA}}$$

$$\mathrm{III} = \mathbb{E}^Q\left[D_0(\tau^B)V^*(\tau^B)\mathbb{I}_{\tau^B<T}\mathbb{I}_{\tau^B<\tau^C}\right] + \underbrace{\mathbb{E}^Q\left[D_0(\tau^B)(1-R^B)V^*(\tau^B)^-\mathbb{I}_{\tau^B<T}\mathbb{I}_{\tau^B<\tau^C}\right]}_{\text{DVA}}$$

BCVA & DVA
Bi-lateral CVA derivation 2

Consider single derivative trade with payoff $\Phi(T)$ at maturity T

$$V_{tot} = \mathbb{E}^Q\left[D_0(T)\Phi(T)\mathbb{I}_{\tau_B,\tau_C>T}\right] + \mathbb{E}^Q\left[D_0(\tau^C)V^*(\tau^C)\mathbb{I}_{\tau^C<T}\mathbb{I}_{\tau^C<\tau^B}\right] + \mathbb{E}^Q\left[D_0(\tau^B)V^*(\tau^B)\mathbb{I}_{\tau^B<T}\mathbb{I}_{\tau^B<\tau^C}\right] - \mathrm{BCVA} + \mathrm{DVA}$$

- $V^*(t) \equiv \mathbb{E}_t^Q[D_t(T)\Phi(T)]$
- $\mathbb{E}_0^Q\left[D_0(t)\mathbb{E}_t^Q[D_t(T)\Phi(T)]\right] = \mathbb{E}_0^Q\left[\mathbb{E}_t^Q[D_0(t)D_t(T)\Phi(T)]\right] \overset{\text{Tower Law}}{=} \mathbb{E}_0^Q[D_0(T)\Phi(T)]$
- $\mathbb{I}_{\tau_B,\tau_C>T} + \mathbb{I}_{\tau^C<T}\mathbb{I}_{\tau^C<\tau^B} + \mathbb{I}_{\tau^B<T}\mathbb{I}_{\tau^B<\tau^C} = 1$

$$\boxed{\begin{aligned} V_{tot} &= \mathbb{E}_0^Q[D_0(T)\Phi(T)] - \mathrm{BCVA} + \mathrm{DVA} \\ &= V_0^* - \mathrm{BCVA} + \mathrm{DVA} \end{aligned}}$$

FVA
FVA Derivation

Valuation from the perspective of the Shareholder

- In default the value of any asset on the balance sheet is worth $R^B V^*(\tau^B)$

$$V^*(t)^+ \equiv \max(V^*(t),0)$$
$$V^*(t)^- \equiv -\min(V^*(t),0)$$
$$V^*(t) = V^*(t)^+ - V^*(t)^-$$

$$V_{tot} = \underbrace{\mathbb{E}^Q\left[D_0(T)\Phi(T)\mathbb{I}_{\tau_B,\tau_C>T}\right]}_{\text{I: no default}} + \underbrace{\mathbb{E}^Q\left[D_0(\tau^C)\left(R^C V^*(\tau^C)^+ - V^*(\tau^C)^-\right)\mathbb{I}_{\tau^C<T}\mathbb{I}_{\tau^C<\tau^B}\right]}_{\text{II: C default}}$$

$$+ \underbrace{\mathbb{E}^Q\left[D_0(\tau^B)\left(R^B V^*(\tau^B)^+ - R^B V^*(\tau^B)^-\right)\mathbb{I}_{\tau^B<T}\mathbb{I}_{\tau^B<\tau^C}\right]}_{\text{III: B default}}$$

$$\text{II} = \mathbb{E}^Q\left[D_0(\tau^C)V^*(\tau^C)\mathbb{I}_{\tau^C<T}\mathbb{I}_{\tau^C<\tau^B}\right] - \underbrace{\mathbb{E}^Q\left[D_0(\tau^C)(1-R^C)V^*(\tau^C)^+\mathbb{I}_{\tau^C<T}\mathbb{I}_{\tau^C<\tau^B}\right]}_{\text{BCVA}}$$

$$\text{III} = \mathbb{E}^Q\left[D_0(\tau^B)V^*(\tau^B)\mathbb{I}_{\tau^B<T}\mathbb{I}_{\tau^B<\tau^C}\right] - \underbrace{\mathbb{E}^Q\left[D_0(\tau^B)(1-R^B)V^*(\tau^B)\,\mathbb{I}_{\tau^B<T}\mathbb{I}_{\tau^B<\tau^C}\right]}_{\text{FVA}}$$

$$\boxed{V_{tot} = V_0^* - \text{BCVA} - \text{FVA}}$$

B Shareholder pays/receives	Time	$V^*>0$	$V^*<0$
I: No default	T	$\Phi(T)$	
II: C defaults first	τ^C	$R^C \cdot V^*(\tau^C)^+$	$-V^*(\tau^C)^-$
III: B defaults first	τ^B	$R^B \cdot V^*(\tau^B)^+$	$-R^B \cdot V^*(\tau^B)^-$

FVA
FVA Decomposition

$$V^*(t)^+ \equiv \max(V^*(t), 0)$$
$$V^*(t)^- \equiv -\min(V^*(t), 0)$$
$$V^*(t) = V^*(t)^+ - V^*(t)^-$$

FVA can positive or negative

- FVA can be split into a cost and a benefit

$$\mathrm{FVA} = \mathbb{E}^Q\left[D_0(\tau^B)(1-R^B)V^*(\tau^B)\,\mathbb{I}_{\tau^B<T}\mathbb{I}_{\tau^B<\tau^C}\right]$$

$$= \underbrace{\mathbb{E}^Q\left[D_0(\tau^B)(1-R^B)V^*(\tau^B)^+\mathbb{I}_{\tau^B<T}\mathbb{I}_{\tau^B<\tau^C}\right]}_{\text{FCA (funding cost adjustment)}}$$

$$- \underbrace{\mathbb{E}^Q\left[D_0(\tau^B)(1-R^B)V^*(\tau^B)^-\mathbb{I}_{\tau^B<T}\mathbb{I}_{\tau^B<\tau^C}\right]}_{\text{FBA (funding benefit adjustment)}}$$

- FVA = FCA - FBA
- The funding benefit has same form as DVA: FBA = DVA

FVA Hack for Intuition

What is the cost of funding a derivatives trade

- Floating rate note with riskless counterparty:
- Bank B will make running loss of $(r^B - r^*) = s^B$
- Where s^B is the funding spread of B and N is notional
- Cost is incurred while B is alive
- Total cost is:

Cost incurred to maturity

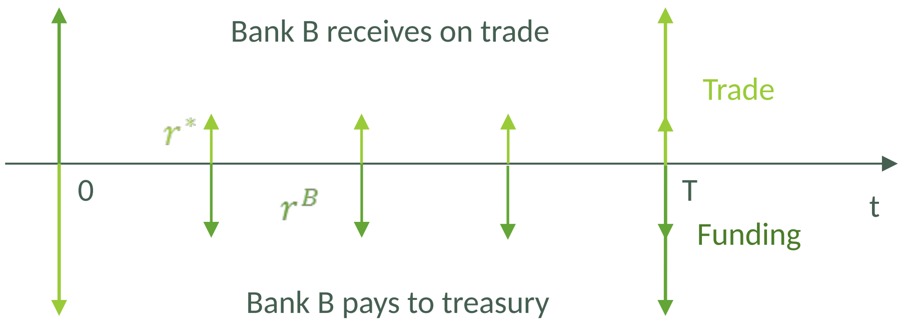
$$\int_0^T e^{-r^* t} Q^B(t) s^B(t)\, dt$$

Discount factor

B Survival Prob

Funding spread

$$s^B = \lambda^B (1 - R^B)$$



Floating rate note:

- Buy 1y floating rate note for \$1 from risk free counterparty
- Get paid quarterly coupons of risk-free rate r^*
- Get paid notional of \$1 at maturity
- To buy the note need to borrow \$1 from treasury
- Treasury will charge funding rate r^B until I pay back at maturity

$$\text{FVA} \sim \int_0^T S^B(t) Z(t) V(t)\, dt$$

FVA in Practice

▼ FVA Component Results

	FVA EPE DD	FCA	FBA	FVA	FVA EPC	FVA EPE	FVA DD	FVA Fundi...	Risky FVA	FVAX	FVA ENE	FVA ENC	FVA ENE DD
1	3,109,016.64	1,109.38	-337.25	**1,446.63**	-43,883,132.63	43,884,242.02	46.06	1,400.57	1,126.26	320.37	-27,568,065.24	27,568,402.49	-1,343,040.76

- Generalized FVA.
- FVA have many components; FBA, FCA, FVA DD, FVA-Funding, Risky FVA, etc.
- The formula looks similar.
  - A set of three (r_1, r_2, r_3) interest rates.
  - Future Expected MTM or Collateral Profile (Z_u, X_u).
  - Integration over time. Calculated numerically.

Sample Date	EP FWD MTM	EN FWD MTM	NET FWD MTM	EPC
02/28/2025	0.00	124,115,423.80	**-124,115,423.80**	0.00
03/01/2025	0.00	124,129,575.24	**-124,129,575.24**	0.00
03/12/2025	0.00	124,284,950.47	**-124,284,950.47**	0.00
03/19/2025	0.00	124,401,679.99	**-124,401,679.99**	0.00
04/02/2025	11,831.17	124,558,271.79	**-124,546,440.62**	11,831.17
04/23/2025	210,953.55	115,036,585.07	**-114,825,631.52**	213,995.99
05/21/2025	977,831.68	116,171,079.93	**-115,193,248.24**	983,952.53
06/18/2025	2,294,772.48	117,647,438.21	**-115,352,665.73**	2,301,349.60
07/16/2025	3,985,463.55	119,747,736.59	**-115,762,273.04**	3,982,543.22
08/13/2025	5,052,621.14	121,435,306.54	**-116,382,685.40**	5,043,266.57
09/10/2025	7,045,179.12	123,913,439.31	**-116,868,260.19**	7,064,202.27
10/08/2025	8,373,379.72	125,743,759.28	**-117,370,379.56**	8,360,566.82
11/05/2025	12,419,969.48	120,415,267.17	**-107,995,297.70**	12,383,425.58

$$E\left[\int_0^\infty e^{\int_0^u r_3(s)ds}\left(r_1(u)-r_2(u)\right)Z_u du\right]$$

$$E\left[\int_0^\infty e^{\int_0^u r_3(s)ds}\left(r_1(u)-r_2(u)\right)X_u du\right]$$

FVA in Practice(FVA_EPE_DD)

$$E\left[\int_0^{\infty} e^{\int_0^u r_3(s)ds}\left(r_1(u)-r_2(u)\right)Z_u du\right]$$

FVA EPE DD
3,109,016.64

Sample Date	EP FWD MTM
02/28/2025	0.00
03/01/2025	0.00
03/12/2025	0.00
03/19/2025	0.00
04/02/2025	11,831.17
04/23/2025	210,953.55
05/21/2025	977,831.68
06/18/2025	2,294,772.48
07/16/2025	3,985,463.55
08/13/2025	5,052,621.14
09/10/2025	7,045,179.12
10/08/2025	8,373,379.72
11/05/2025	12,419,969.48
12/03/2025	14,946,604.99
12/31/2025	18,031,305.53
01/11/2026	19,123,787.58
01/28/2026	20,588,007.04
04/22/2026	28,941,473.18
08/26/2026	35,583,802.86
06/16/2027	54,708,199.92
04/05/2028	73,716,360.92

- EP FWD MTM Profile.
- Libor (Sofr), Valuation/Collateral rate, Funding rate.
- Estimate using Numerical approximation.

$$0 = t_0 < t_1 < t_2 \cdots < t_n$$

$$\sum_{i\geq 1} E\left[-Z_{t_{i-1}} M_3(t_{i-1})^{-1}\left(\frac{M_1(t_{i-1})}{M_2(t_{i-1})}\right)\left(\frac{M_2(t_i)}{M_1(t_i)}-\frac{M_2(t_{i-1})}{M_1(t_{i-1})}\right)\right]$$

FVA in Practice(FVA_EPE_DD)

Trapezoidal Rule:

SUM | =-1/2*((C4*(G4/H4)*E4)+I4*(M4/N4)*K4)*(H4/G4-N4/M4)

	C	E	G	H	I	K	M	N	T	V	X
1	disc_i_1	profileValues[i + 1]	r2_i_1	r1_i_1	disc_i	profileValues[i]	r2_i	r1_i	Date	CalculationFormula	Running Sum
2	1.0005	0	1.0005	1.00048	1.00062	-	1.00063	1.0006		0	
3	1.00037	0	1.00038	1.00036	1.0005	-	1.0005	1.00048	2/28/2025	0	
4	0.999006	0	0.999005	0.99904	1.00037	-	1.00038	1.00036	3/1/2025	H4/G4-N4/M4)	
5	0.998139	0	0.998149	0.998202	0.999006	-	0.999005	0.99904	3/12/2025	0	
6	0.996406	11,831	0.996464	0.996533	0.998139	-	0.998149	0.998202	3/19/2025	0	0
7	0.993807	210,954	0.993979	0.994044	0.996406	11,831	0.996464	0.996533	4/2/2025	0	0
8	0.990347	977,832	0.990697	0.990748	0.993807	210,954	0.993979	0.994044	4/23/2025	8	9
9	0.986939	2,294,772	0.987475	0.987521	0.990347	977,832	0.990697	0.990748	5/21/2025	8	16
10	0.983595	3,985,464	0.984331	0.984381	0.986939	2,294,772	0.987475	0.987521	6/18/2025	-13	3
11	0.980128	5,052,621	0.981237	0.981295	0.983595	3,985,464	0.984331	0.984381	7/16/2025	-37	-33
12	0.976452	7,045,179	0.978189	0.978256	0.980128	5,052,621	0.981237	0.981295	8/13/2025	-56	-89
13	0.973137	8,373,380	0.975224	0.975296	0.976452	7,045,179	0.978189	0.978256	9/10/2025	-40	-129
14	0.970015	12,419,969	0.972323	0.972401	0.973137	8,373,380	0.975224	0.975296	10/8/2025	-65	-194

	profileValues[i + 1]	r2_i_1	r1_i_1	disc_i	profileValues[i]	r2_i	r1_i	Date	CalculationFormula	Running Sum
4	471,059,196	0.400233	0.391945	0.315766	471,059,196	0.424035	0.417329	12/4/2047	709,567	3,109,035